CHURRO: Critical Design Review



CRANSEDS

RACE 2 SPACE

Cranfield University

~~25/04/25~~

02/05/25

1 Introduction

This year the CranSEDS propulsion team have set the goal of developing a flight capable hybrid rocket system. In this way the team shall build upon the progress made at CranSEDS in previous years. The rocket shall nominally be able to lift a 15kg upper section to an altitude of 3000m. These goals have been based upon the requirements of EUROC. Although the team shall not be competing in EUROC 2025, it will be integrating the engine into a rocket for a launch in August 2025.

This year's rocket design has taken two projects as its main source of inspiration. The combustion section is building on the previous work completed at CranSEDS. For the fluid and tank system the team is basing the design on Half-Cat Rocketry's Mojave Sphinx project(1). Half-Cat Rocketry is an American amateur rocketry group which was set up with the goal of reducing the barrier of entry for liquid rocketry. Mojave Sphinx is an open-source design that they have extensively documented (Mojave Sphinx – Half Cat Rocketry). This design includes well documented and tested methods for constructing pressure vessels and fluid delivery systems with facilities accessible to amateurs. Not only have Half-Cat Rocketry ran an extensive test regime, but many amateur teams worldwide have also successfully replicated and launched this design.

1.1 Test Goals

As our overall objective is to use this engine in a launch, we want to have tested the entire system in as near as possible a flight configuration. These tests will verify the accuracy of our calculated thrust and mass flows. They will also test the functionality of our fluid control and

2 System Overview

Max Thrust (N)	Average Thrust (N)	Burn Time (s)	Oxidiser Mass (kg)	Fuel Mass (kg)	Overall Length
2123	1472	8	6	1.5	2.2 meters

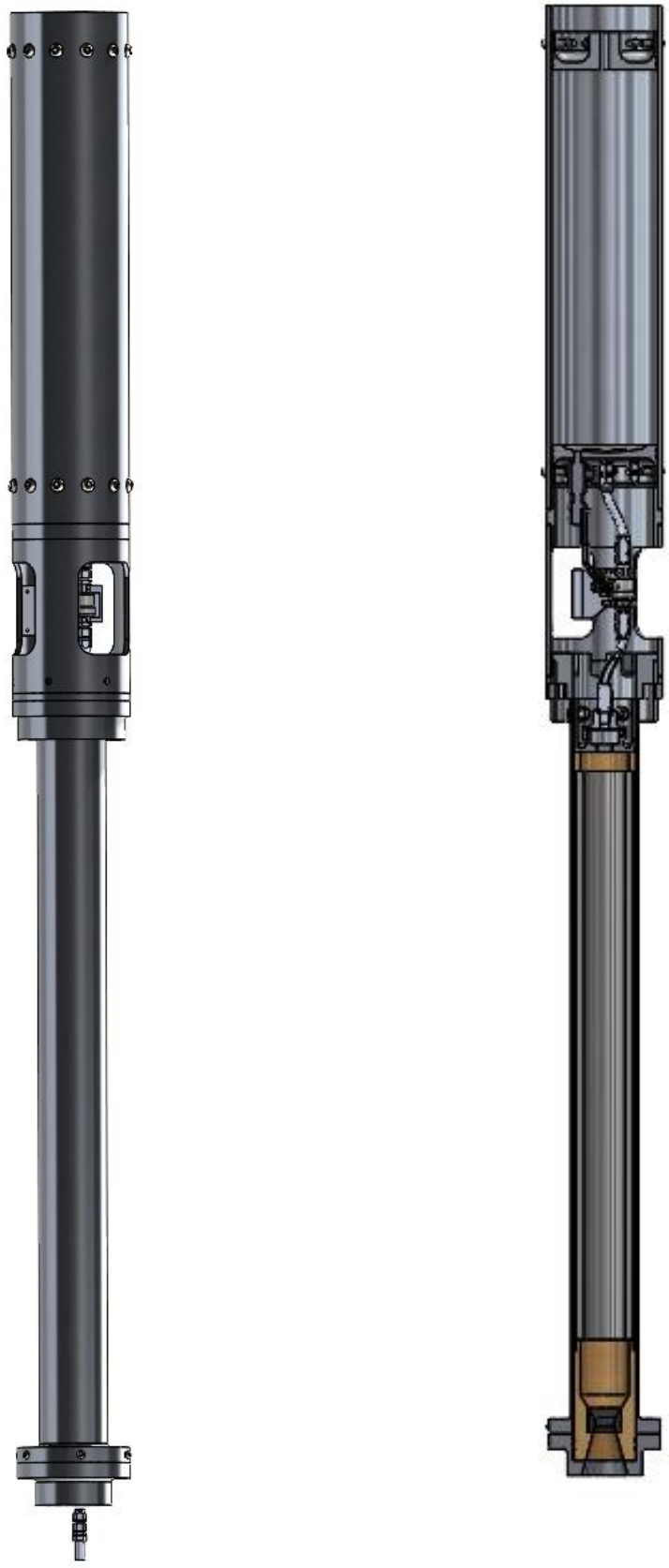


Figure 1: Isometric and section view of engine

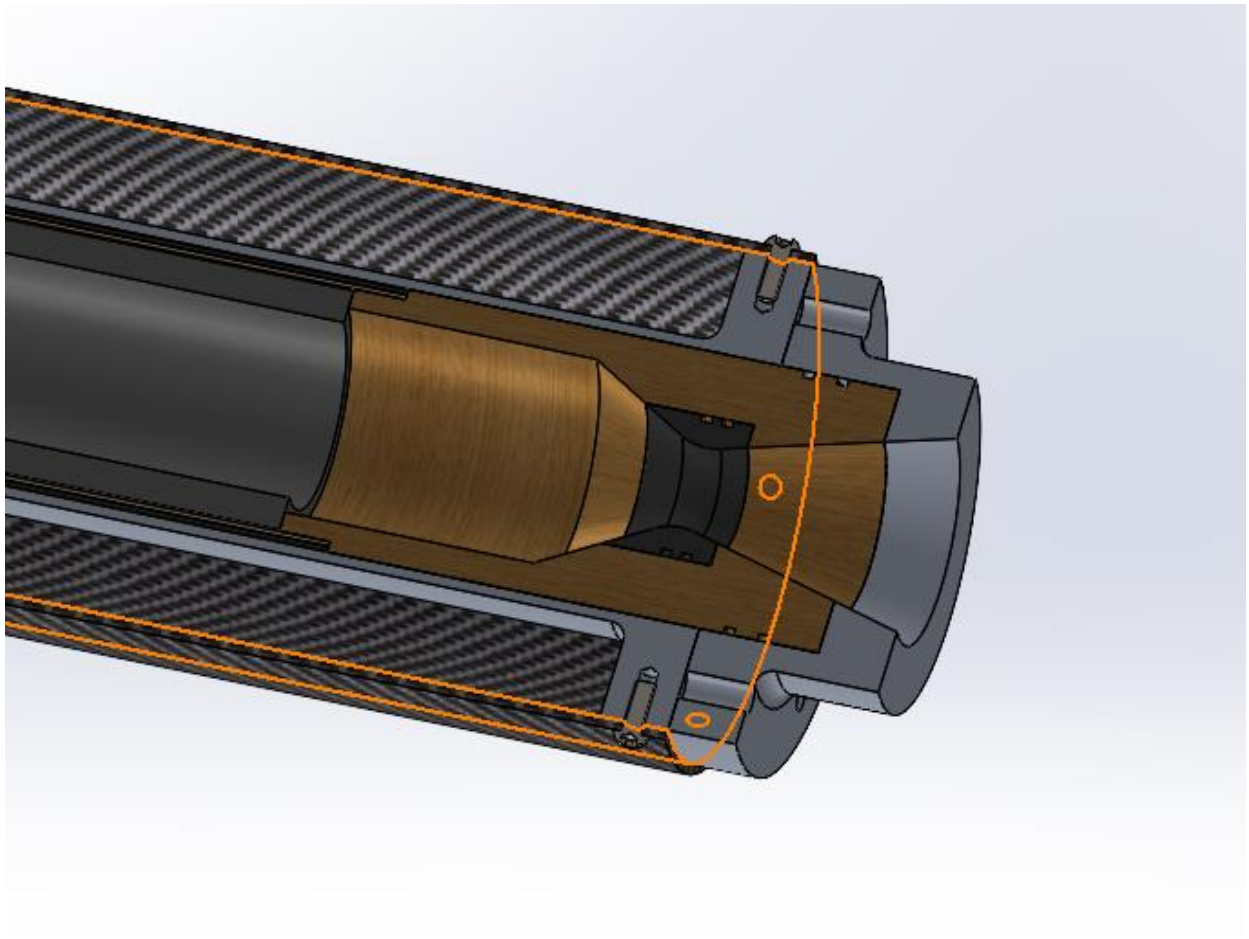


Figure 2: Highlight of nozzle section

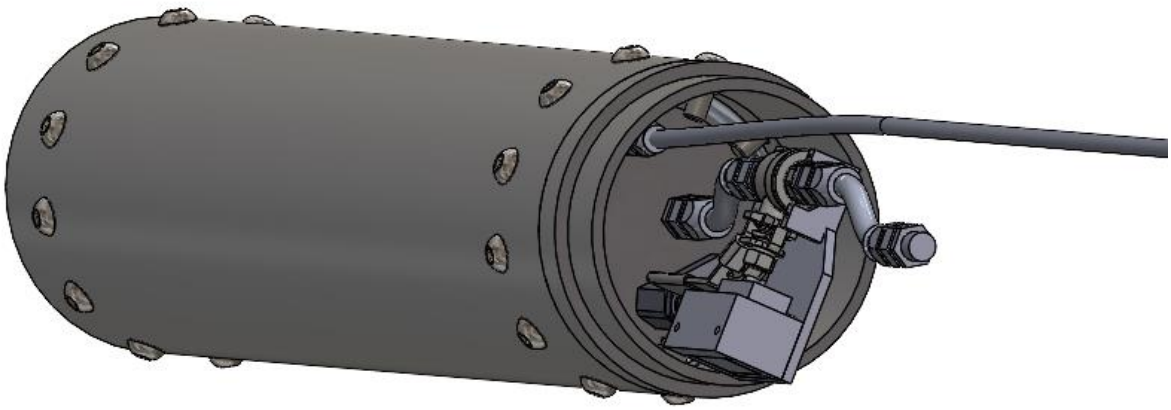


Figure 3:Lower Tank Bulkhead

3 Project Timeline and Budget

A detailed Bill of Materials and Gantt can be found in Appendices 3 and 4.

We have been fortunate enough to have the support of the Centre for Propulsion and Thermal Power Engineering who have graciously agreed to fund the project. This includes the procurement and manufacturing costs.

Manufacturing will mainly take place in campus facilities and a dialogue has begun with the relevant technicians.

4 Fluid System

4.1 Tank

4.1.1 Structural Analysis and Design

This analysis assesses the mechanical integrity of the pressure vessel designed to store nitrous oxide (N_2O) at a maximum expected operating pressure (MEOP) of 11 MPa. The vessel is constructed with an aluminium 6061-T6 liner and utilises 12 Grade 4.6 steel bolts for end cap attachment. Stress and safety factor evaluations were performed to validate the design against potential failure modes.

Pressure Vessel Factor of Safety	3
Proof Pressure (MPa)	11
Bolt Shear Stress (MPa)	255
Bolt Shear Factor of Safety	1.76
Bolt Tear Out Stress (MPa)	24.6
Bolt Tear Out Factor of Safety	8.38
Casing Tensile Stress (MPa)	78.36
Casing Tensile Factor of Safety	3.06
Bearing Stress (MPa)	221.3
Bearing Factor of Safety	1.74

Based on the above analysis, the N_2O tank and fastener assembly demonstrate acceptable performance under the maximum expected operating pressure of 11 MPa. All key stress modes, including bolt shear, tear-out, casing tensile, and bearing stresses, have been evaluated. While the design maintains adequate safety margins overall, the relatively lower safety factors observed in bolt shear and bearing stress indicate potential weak points that may be more prone to localised failure, such as bolt pull-out or flange damage; these are preferred failure points rather than a complete structural failure of the tank. These areas want

closer inspection and possible reinforcement to ensure reliability under dynamic loading or long-term use.

4.2 Valves and Piping

4.2.1 Onboard Fill Lines

The oxidiser tank shall be filled through the bottom tank bulkhead. A feed line will be run through the combustion chamber flanges to the bottom of the vehicle, exiting next to the nozzle via a quick connect.

The feed line will use AN4 steel braided PTFE hose, AN4 straight fittings, and AN4 to 1/8 " BSPP adaptors. With one end connected to the bottom bulkhead and the other to a quick connector. All components are rated to more than the 55 bar maximum pressure expected with most being rated significantly higher.

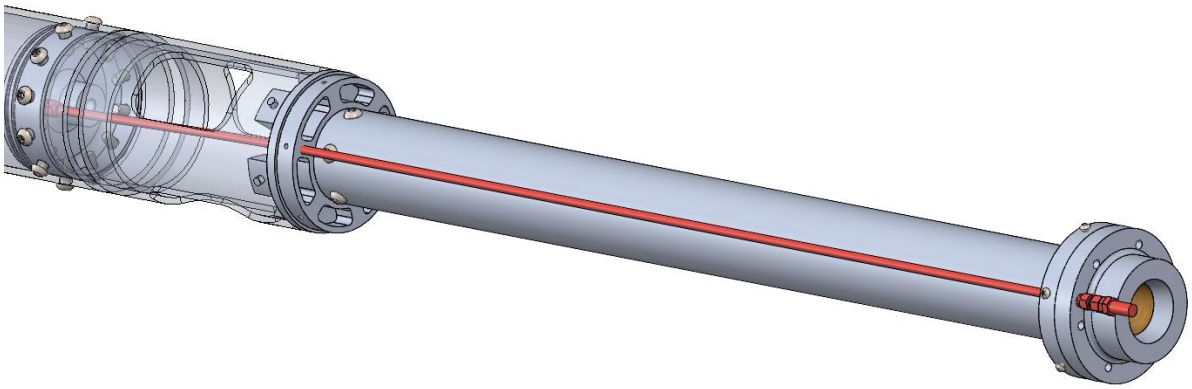


Figure 2:Internal Fill Line

Using this system, the oxidiser tank can be filled within 2 minutes. This feed rate is limited by the static vent rather than the ground feed system.

The quick connector is typically used for hydraulic systems which means it self-seals upon disconnection and is rated to several times the pressure (350 bar) of our system (55 bar). A slight modification to the quick connect is required to disengage the lock upon connection. The vehicle shall act as the holding force when vertical on the launch rail. The quick connect will not be used for Race 2 Space however it is necessary for the flight later this year.



Figure 3: Hydraulic QD

4.2.2 Main Oxidiser Flow Assembly

The main oxidiser flow assembly is responsible for controlling the flow of nitrous oxide from the onboard tank to the injector. The assembly centres around a servo-

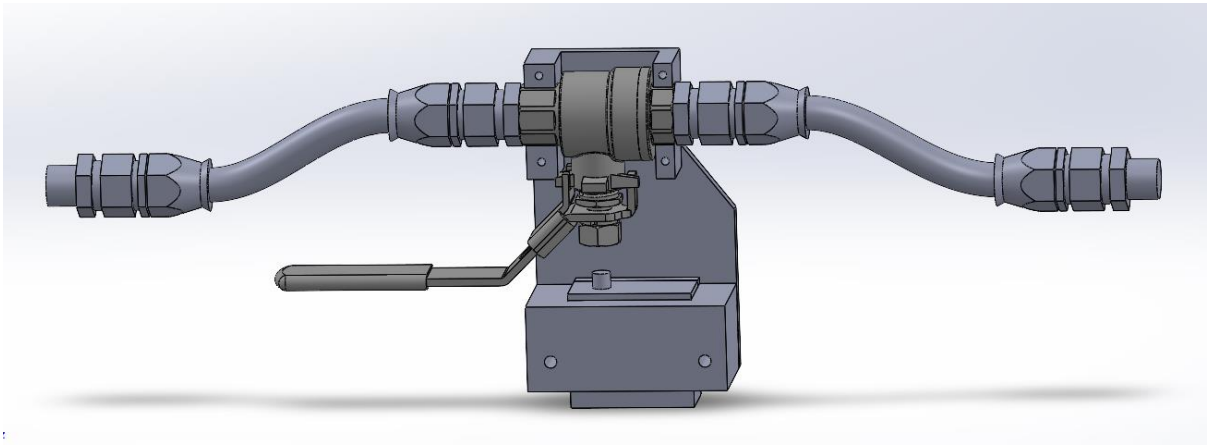


Figure 4: Main Fluid Line

actuated ball valve controlled by our engine controller. The ball valve is connected using AN6 steel braided PTFE pipework to give us flexibility during assembly.

The ball valve used is a 2-way stainless stell ball valve with PTFE seals, rated up to 63 bar. This valve has a bore of 11.5mm and a Kv of 6. Injector calculations suggest that a Kv of 2 is required for our system, so this valve provides significant overhead. This is a manually operated valve which will be modified to actuate using a servo motor, hence a 3D printed mount is required. This will follow similar systems found online such as Half Cat.

The lines to and from the valve are identical, taking a 1/4" BSPP thread from the bulkheads of our tank and injector, converting into AN6 for the pipes and back to 1/4" BSPP for the valve. Steel braided PTFE lines were chosen due to their ease of

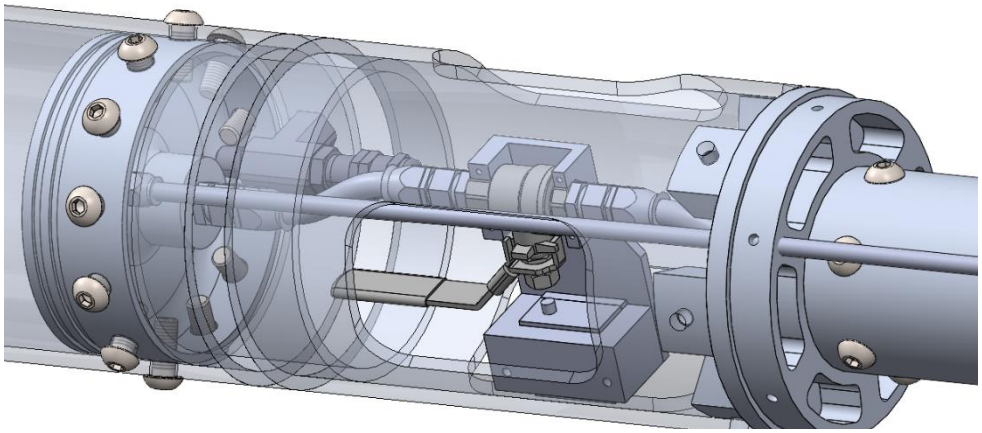


Figure 5: Fluid Line in Coupler

manufacture, flexibility, and excellent pressure resistance.

4.2.3 Pressure Relief Valve

The tank shall have a pressure relief valve fitted as a safety measure. We shall be using a Swagelok stainless steel high pressure proportional relief valve. This valve is spring loaded and adjustable, letting us set the desired relief pressure. In practice this permits us to set the relief pressure close to our operating pressure protecting other components in our system.



Figure 6: Pressure Relief Valve

The relief valve was selected over a burst disk for two main reasons:

- We do not lose all the nitrous oxide in the tank should we over pressurise the tank, instead the excess pressure is released and brought back to a safe level letting operations continue.
- Burst disks are single use custom components that we as a team did not feel confident in manufacturing consistently ourselves. With the Swagelok valve, there is little doubt as to the effectiveness of the safety device.

The relief valve has a mass flow rate greater than the oxidiser feed line ensuring that even if the feed system continues to pressurise the tank, the relief valve can still reduce the pressure within the system.

4.2.4 Static Vent

The oxidiser tank also features a static vent of 0.8mm which serves two purposes. First it acts as a safety device to depressurise the tank should control to the motor be lost. This process should take approximately 10mins to go from fully pressurised to atmospheric pressure. The second function is to vent nitrous

vapour whilst filling the tank. Without a vent, the initial nitrous oxide liquid would vaporise pressurising the tank and preventing filling beyond 10-20%.

The vent diameter was selected based on the associated fill time and tank depressurisation time. These were determined using the following equations:

$$A_{vent} = \pi r^2$$

Gas Density @ 15 °C taken from NIST – 1.3E+2

Gamma (Cp/Cv) @ 15 °C taken from NIST – 2.6

$$\dot{m}_{vent} = \frac{A_{vent} \cdot P_{tank}}{\sqrt{T_{tank}} \cdot \sqrt{\frac{\gamma}{R}} * \frac{\gamma + 1}{2}^{\frac{1-\gamma}{2(\gamma-1)}}} = 9.9E-03\ kg/s$$

$$\dot{V}_{vent} = \frac{\dot{m}_{vent}}{\rho_{gas}} = 7.4E-05\ m^3/s$$

The estimated tank fill time is given by:

$$T_{fill} = \frac{V_{ox}}{\dot{V}_{vent}} = 112\ s$$

The estimated tank depressurisation time is given by:

$$t = \frac{m_{ox}}{\dot{m}_{vent}} = 606s\ \approx 10\ min$$

The static vent places a limit on the time between the end of filling operations and ignition of the vehicle. This is unlikely to be an issue for static firing however must be verified before flight.

4.3 Ground Fill Station

The ground fill station is responsible for controlling the filling and ignition sequences. It will house electronics near the pad and will control the flow of nitrous during filling and the flow of oxygen during ignition.

4.3.1 Ground Fluids

The fluid component of the GFS consists of two supply bottles (Nitrous Oxide and Oxygen) with regulators, two servo-actuated ball valves, AN4 braided steel PTFE lines for the Nitrous Oxide, nylon tube for the oxygen supply, and the female quick disconnect (matching fill line).

The oxygen and nitrous lines use identical servo-actuated valves to the one found onboard the vehicle. This is to reduce the number of dissimilar components, simplifying procurement and integration. Since neither line requires the same flow rate, the lines fed into each are reduced in diameter compared to the onboard system.

The lines between the supply bottles and the valves will be kept as short as possible. The lines to the vehicle shall be somewhat longer to distance the supply bottles from the rocket motor. Due to the use of AN connector and braided steel PTFE lines, the feed lines can be relatively easily modified to suite a launch facility. The feed lines are expected to be between 2 and 5 metres long.

4.3.2 Ignition System

The purpose of the ignition system is to initiate the combustion of the engine. It does this by first igniting oxygen gas which is then used to ignite the actual oxidiser, thereby triggering engine ignition and ultimately lift-off. This system has been designed for operational safety. It consists of an oxygen tank that feeds oxygen through a flexible tube running along the length of at least ten meters and up the rocket's fuselage to the top where the oxidiser shower head lies. Alongside this, a set of wires connected to a high-voltage generator is also routed up the fuselage. The wires are fastened to the tube, largely unexposed with exception of 2 cm of stripped wire at the end of the tube to facilitate an electrical arc.

When the ignition system is activated, oxygen flows through the flexible tube, while an electrical signal is simultaneously sent to the high-voltage generator. The resulting potential difference between the two poles creates a spark that ignites the oxygen. Once the rocket's oxidiser begins to flow, the ignited oxygen triggers the combustion of the oxidiser, initiating engine ignition and lift-off. Both the oxygen tank and the high-voltage generator are electronically controlled via a control system, with servo-controlled ball valves regulating fluid flow. Figure 9 bellow illustrates the ignition system.

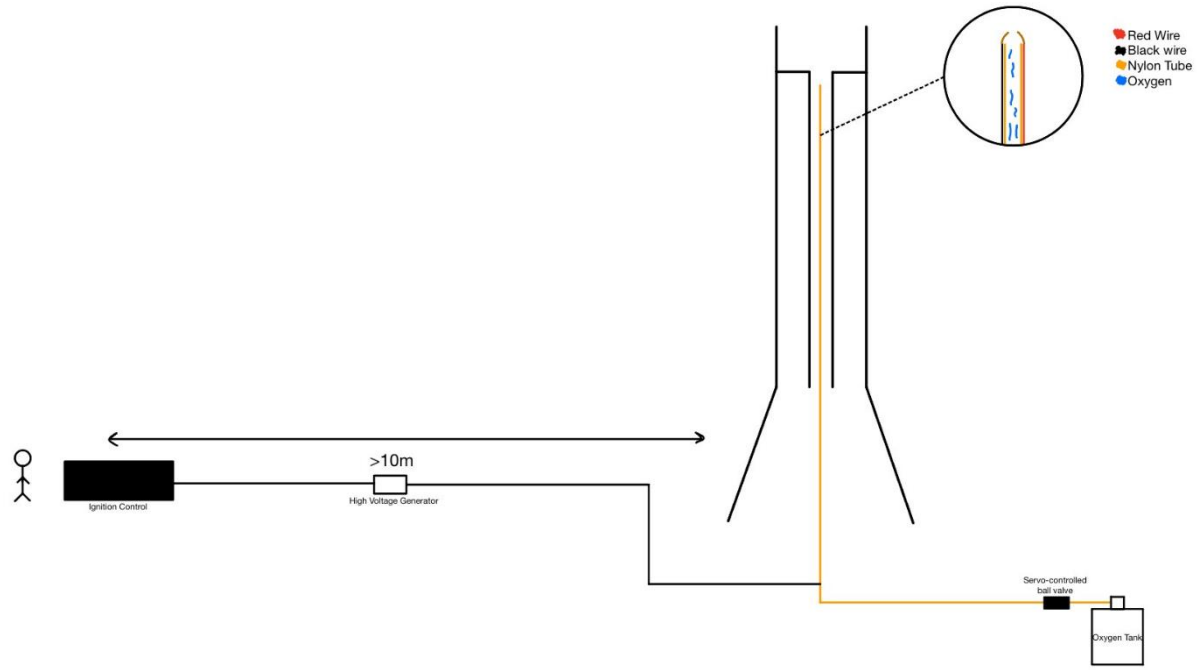


Figure 7: Ignition System

5 Combustion Chamber

5.1 Fuel Grain

The chosen fuel geometry was a Bates grain configuration, as it is the most extensively studied form with the greatest number of references available, in addition to being relatively simple to manufacture. To size the grain, an initial ideal rocket calculation was performed for a 30 kg vehicle targeting an apogee of 3 kilometres. Using the HDPE density of 950 kg/m³ and the required volume, along with references from previous teams' designs, the initial dimensions were estimated.

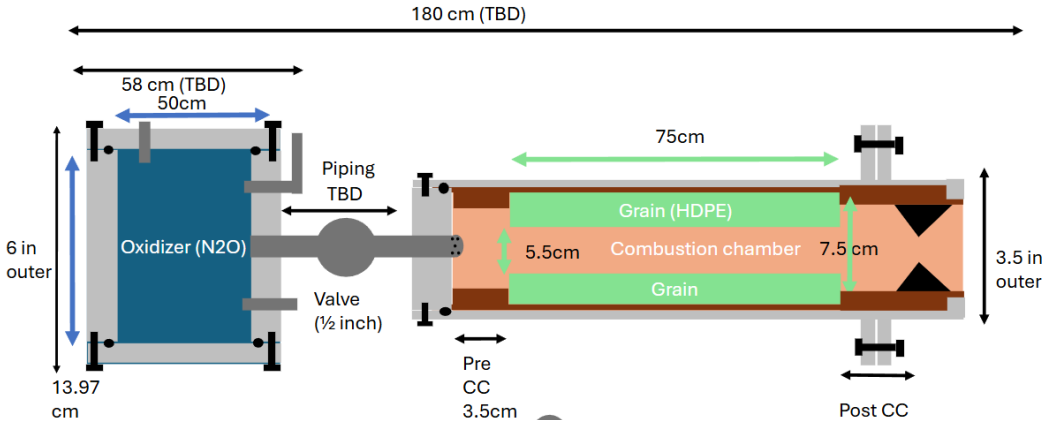


Figure 8: System Layout

Subsequently, the combustion process was simulated using the HRAP (Hybrid Rocket Analysis Program) developed by Robert Nickel at the University of Tennessee, which is based on MATLAB.

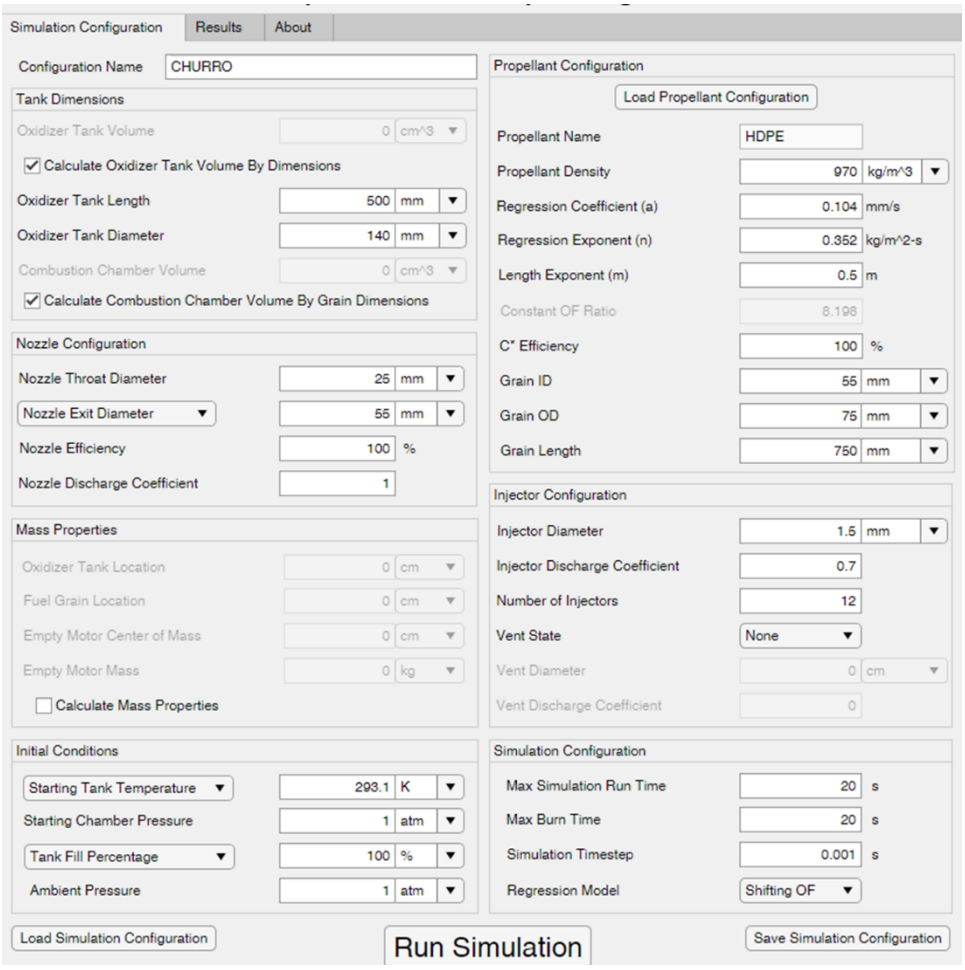


Figure 9: HRAP Inputs

From the initial estimates, the internal diameter was determined to be 5.5 cm, with an external diameter of 7.5 cm and a length of 50 cm, assuming a constant

oxidizer-to-fuel ratio (O/F) of 7. However, when calculations with a variable O/F ratio were performed, it was necessary to extend the grain length to 75 cm to avoid "rich blowout" and "lean" conditions, and to maintain combustion as close as possible to an O/F ratio of 7, which is approximately stoichiometric.

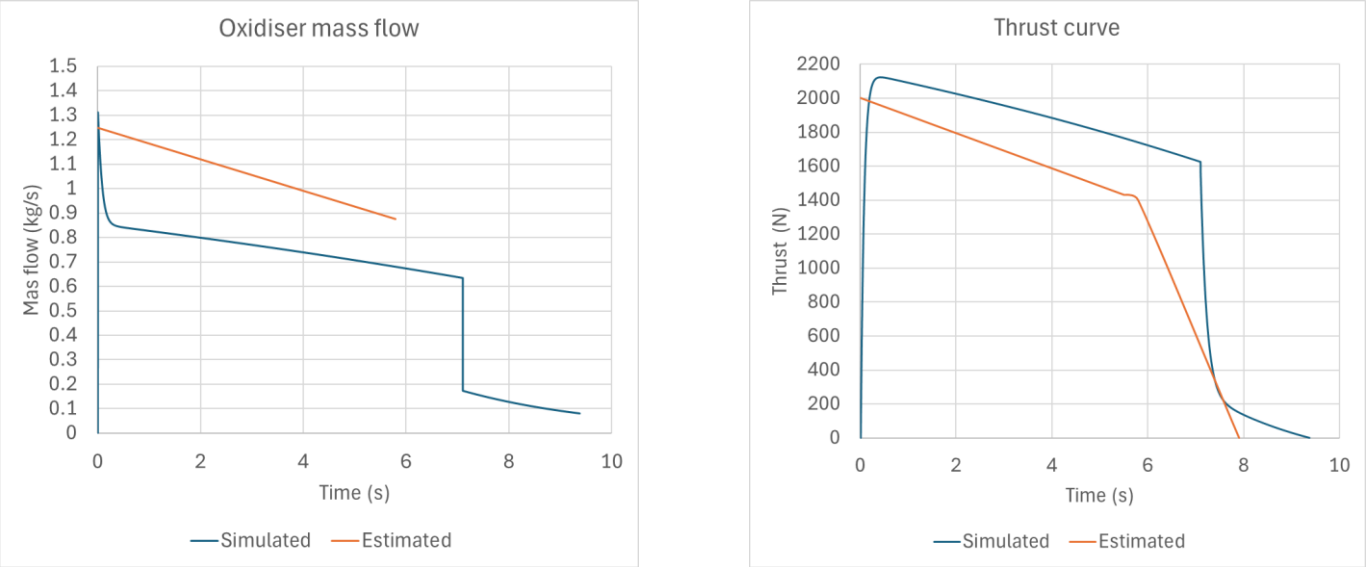
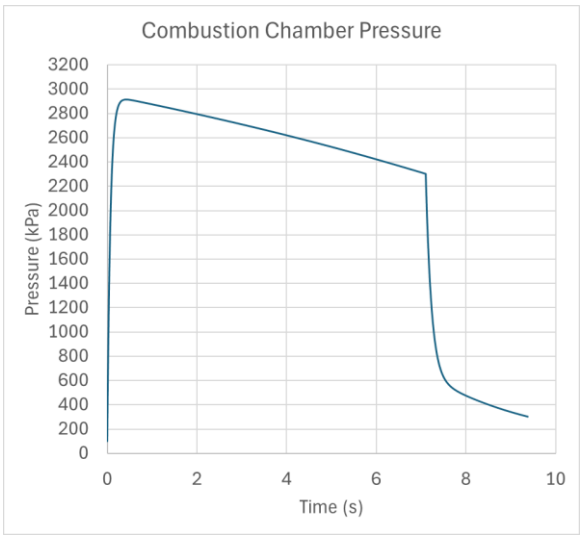
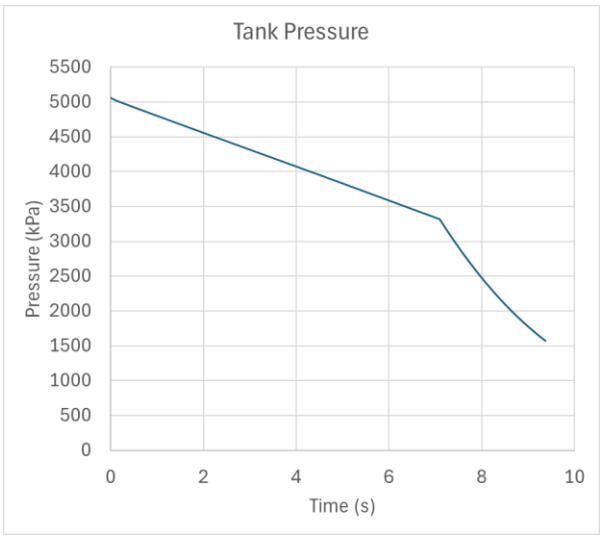


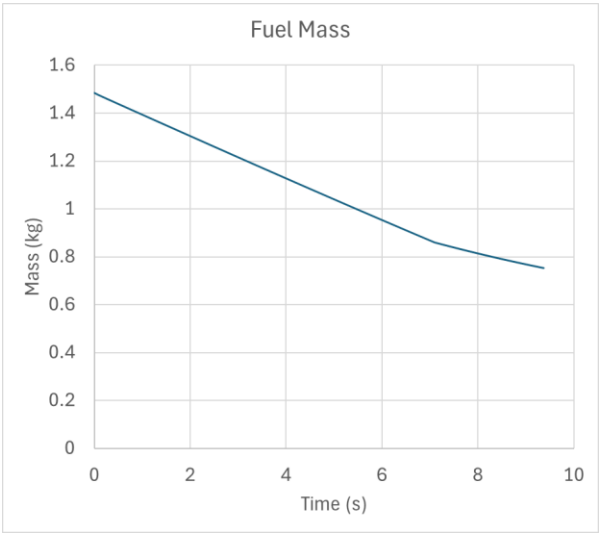
Figure 10: Oxidiser Mass Flow & Thrust Curve from HRAP

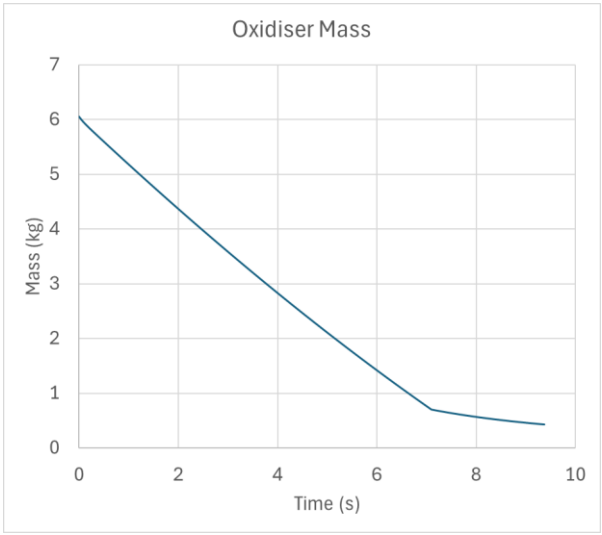
	Time (s)	Thrust (N)	Oxidizer Tank Pressure (kPa)	Combustion Chamber Pressure (kPa)	Injector Pressure Drop (kPa)	Oxidizer Mass (kg)	Fuel Mass (kg)
Initial		0.00	5060.13	101.33	0.00	6.05	1.49
Final (left)	9.37	0.00	1578.06	303.59	0.00	0.43	0.75
Max		2121.78	5060.13	2914.48	0.07	6.05	1.49
Average		1472.45	3739.09	2117.08	-0.13	2.58	1.08

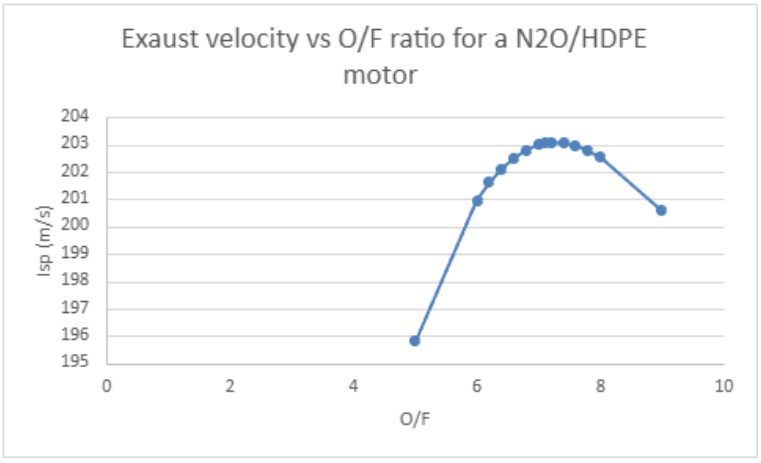
	Oxidizer Mass Flow Rate (kg/s)	Fuel Mass Flow Rate (kg/s)	Exhaust Mass Flow Rate (kg/s)	Oxidizer to Fuel Ratio	Grain ID (cm)	Regression Rate (mm/s)
Initial	0.00	0.00	0.00	0.00	5.50	0.00
Final (left)	0.08	0.04	0.13	1.94	6.56	0.00
Max	1.31	0.10	0.93	12.55	6.56	0.00
Average	0.60	0.08	0.68	7.09	6.10	0.00











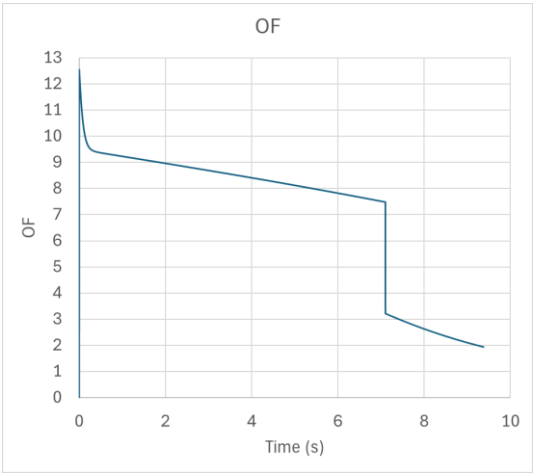


Figure 11: HRAP Outputs

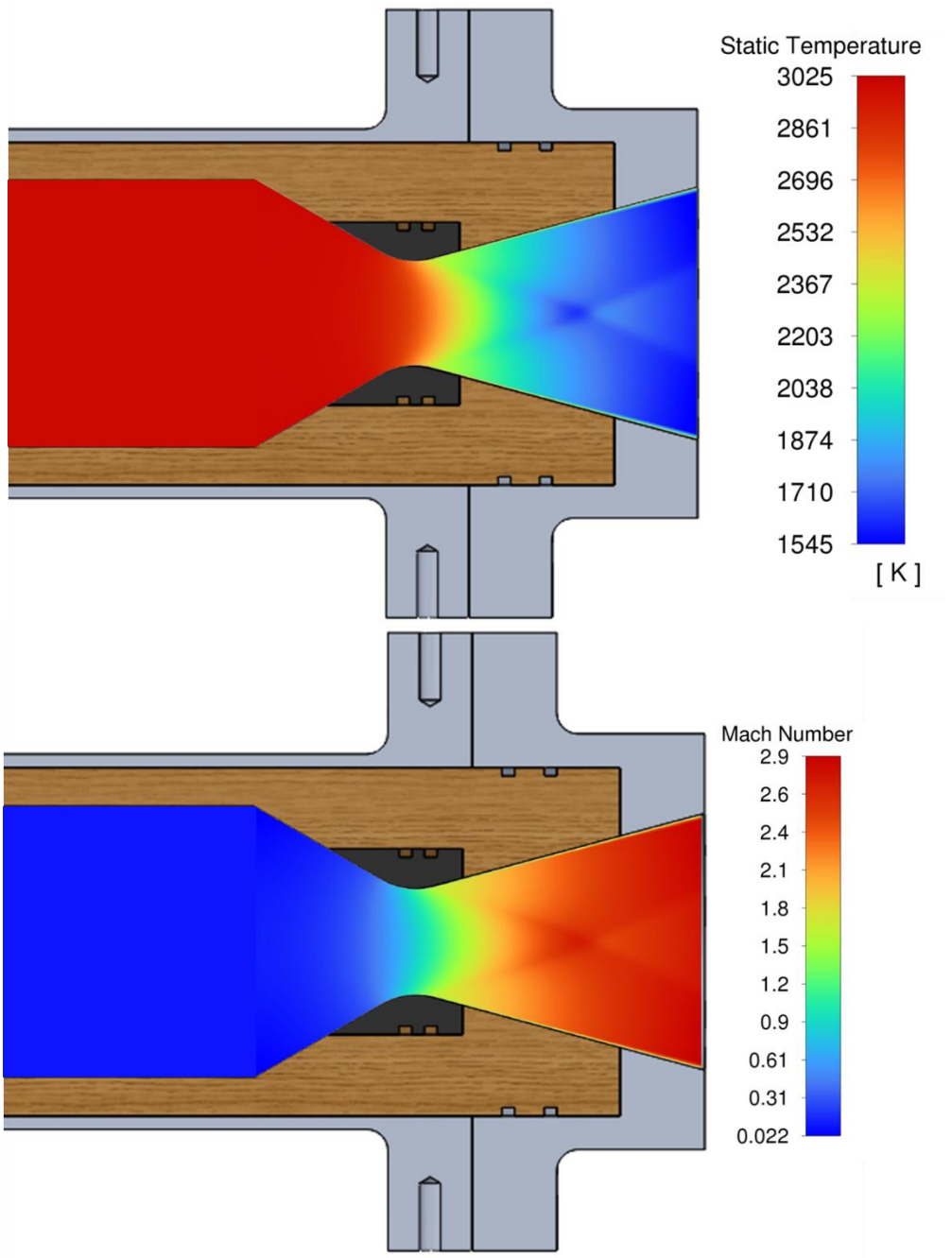


Figure 12: Nozzle CFD

5.2 Structural Analysis and Design

The combustion chamber casing was evaluated at a proof pressure of 6 MPa using aluminium 6061-T6 and Grade 6.8 bolts. This analysis considered key stress conditions such as tensile, shear, tear-out, and bearing stresses to assess the structural integrity of the system. By applying a safety factor of 3, the design aims to ensure that the casing can withstand the maximum expected pressures, maintaining reliability under operational conditions.

Pressure Vessel Factor of Safety	3
Proof Pressure (MPa)	6
Bolt Shear Stress (MPa)	97
Bolt Shear Factor of Safety	4.64
Bolt Tear Out Stress (MPa)	33.4
Bolt Tear Out Factor of Safety	6.17
Casing Tensile Stress (MPa)	48
Casing Tensile Factor of Safety	5.01
Bearing Stress (MPa)	167
Bearing Factor of Safety	2.31

All calculated stress conditions remained within acceptable margins, with the lowest safety factor being 2.31 in bearing stress. While this suggests a localised vulnerability, it does not compromise the overall structural integrity. The design philosophy here embraces controlled failure: engineering for resilience rather than perfection. This approach guarantees that the system is capable of handling extreme conditions, failing in a controlled manner to prevent catastrophic failure and ensure safe operation throughout its lifecycle.

5.3 Heat Transfer Analysis

A detailed explanation of a heat transfer analysis is detailed in Appendix 8. The STAR-CCM+ simulation identified thermal hotspots and mapping transient temperature fields across the chamber geometry. Though it did not include fluid-structure interaction or mechanical stress modelling, the simulation successfully highlighted regions susceptible to overheating, particularly near the nozzle walls. These findings are vital for thermal protection planning and early-stage validation.

Evaluation of **thermal stresses** was not performed in this study, as the chamber is primarily designed to withstand **hoop stresses**, the dominant structural load. Thermal stress effects, though less critical in this context, will be investigated in **future CHT-FEA simulations** to capture full thermo-mechanical behaviour under burn conditions.

Overall, the current study provides a **robust thermal validation** of the chamber design and serves as a practical and reliable basis for **critical design review**. It establishes a computational framework for identifying thermal vulnerabilities and lays the groundwork for future high-fidelity simulations that incorporate stress evaluations to further improve structural resilience.



Figure 13: Temperature within the combustion chamber fluid domain

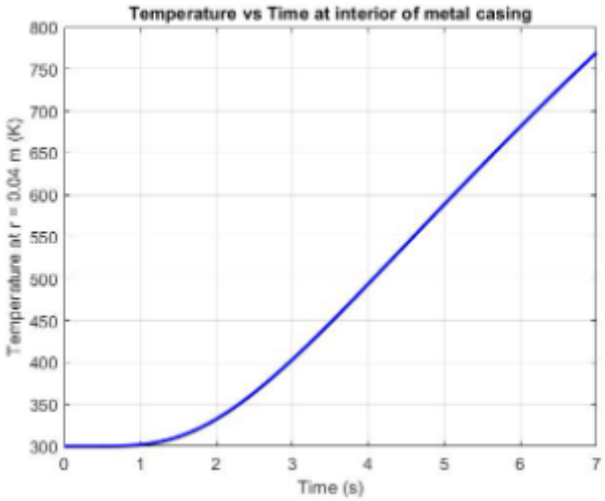


Figure 15:Temperature variation with respect to time at inner radius of metal casing

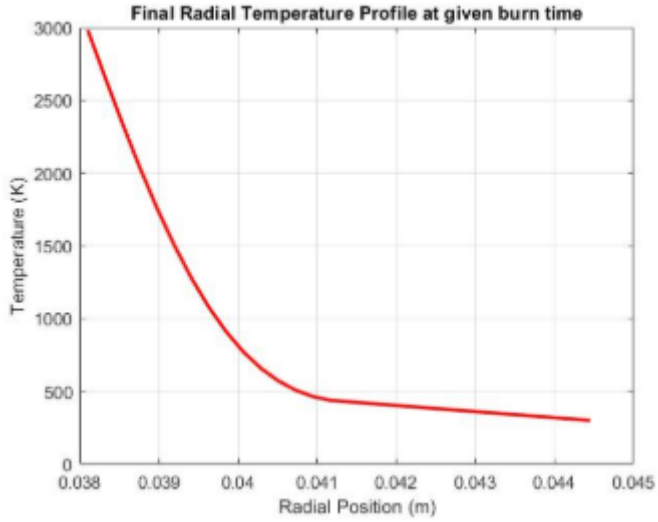


Figure 14:Temperature variation across chamber at burn time.

6 Electronics

6.1 Control Systems

This section provides an in-depth analysis of the design of the launch controller and the ground control system, including a thorough justification for each component selected. It also details the strategic layout of the components and their interfaces to ensure seamless interaction and optimal functionality. The design encompasses simple shelf components to reduce cost and system development time while addressing the specific requirements for safe and reliable pre-launch operations. Diagrams and schematics illustrate the physical arrangement and electrical connections between components.

6.1.1 Launch Controller

To comply with the safety requirements for remote pre-launch operations, such as filling and ignition, Flysky's FS-i6X transmitter is utilized. The FS-i6X operates by toggling various switches that send command signals to the ground system, allowing for the execution of specific functions. This transmitter has been modified to enhance safety during operations. The entire layout of the modified FS-i6X is illustrated in Figure 1.

The FS-i6X features four main operational switches, each designated for different commands ranging from filling to ignition. Each toggle switch is equipped with a safety guard to prevent accidental activation. The toggle switch labelled 'Filling' signals the receiver of the ground control system to activate the servo valve, initiating the filling operation for the nitrous oxide (N2O). Once the filling process is complete, the switch is returned to the 'off' position.

The 'Power' toggle switch is responsible for engaging the relay switch to power the engine controller, effectively pre-arming the rocket and preventing the inadvertent opening of the oxidizer flow valve. The 'Oxidizer' toggle switch opens the servo valve for the oxidizer tank; once the combustion chamber is adequately filled with oxygen, this switch is also set to the 'off' position to close the valve.

To avert the risk of accidental ignition, the ignition toggle switch is secured with a key switch. Activating this switch signals the ground receiver to engage the arc generator relay switch, sending a command to the engine controller to open the N2O flow valve.



Figure 16 - Launch Controller

6.1.2 Ground Control System

The ground control system has been efficiently assembled within a box, as illustrated in Figure 2. This system comprises various essential components, including the Flysky FS-iA10B Receiver, two 14.8-volt LiPo batteries, two screw switches, a buck converter, a relay switch, and an 80-kilovolt arc generator. The Flysky FS-iA10B Receiver is directly compatible with the launch controller, which features a built-in processor and antennas. Two batteries are utilized to ensure safety: one powers the receiver, while the other powers the arc generator. Each battery is connected to a screw switch that allows the ground wire to be opened and closed, preventing accidental activation of the arc generator and ensuring the receiver does not receive any signals inadvertently. The buck converter reduces the 14.8 volts to the 5 volts required for the receiver's operation. The relay switch is responsible for engaging the arc generator operated by the receiver. Finally, the arc generator explicitly generates an arc to ignite the nitrous oxide (N2O) within the combustion chamber. All component connections and interfaces of the ground control system can be seen in Figure 3.

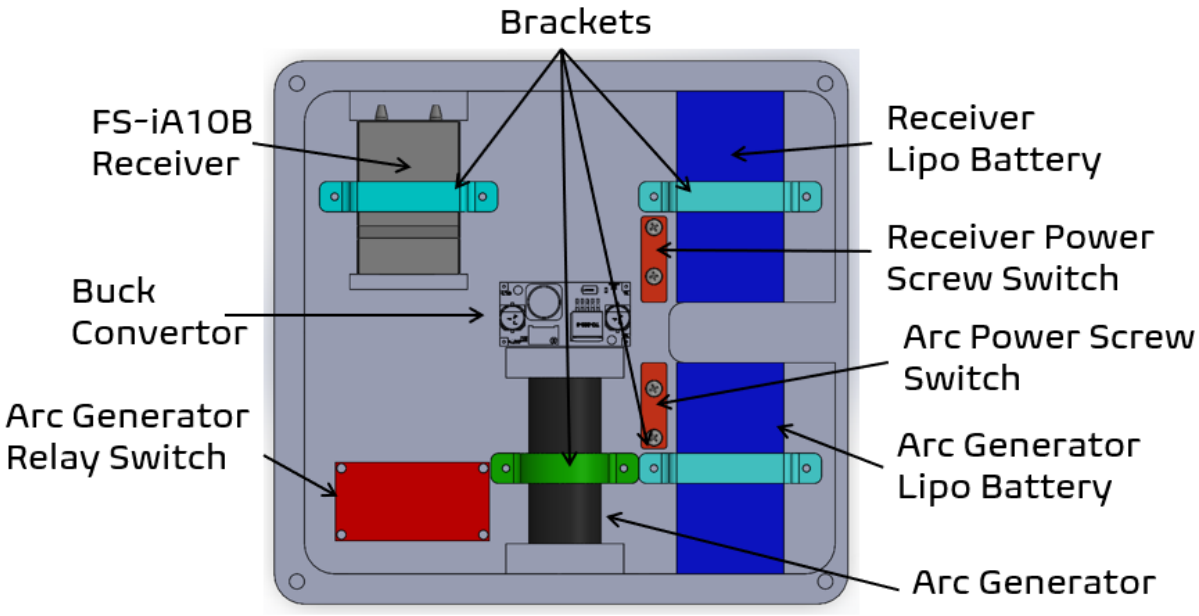


Figure 17 - Ground Control System

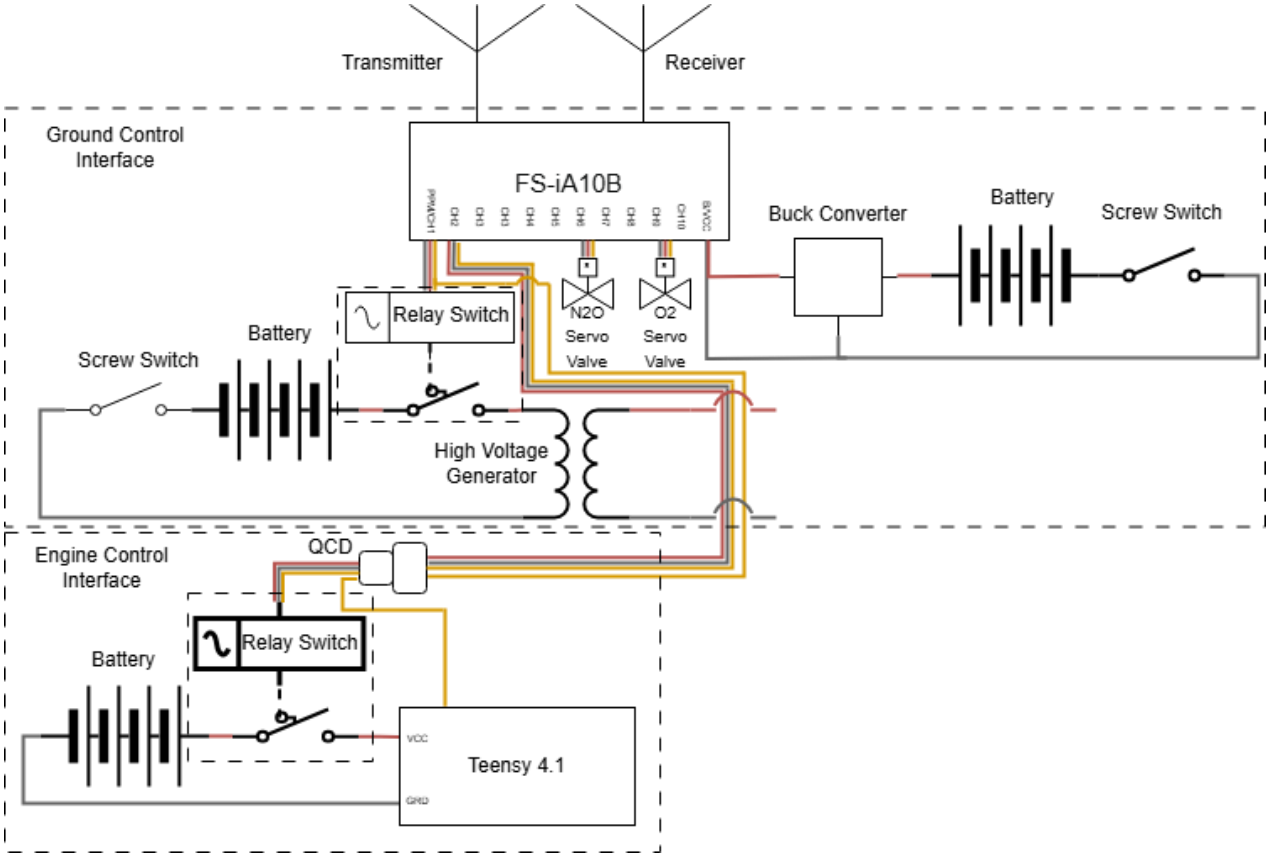


Figure 18 - Electronic Schematic of the Ground Control System

6.2 DAQ Systems

This section details the comprehensive integration of a data acquisition system using the components listed in the provided list of components in Table 6.2.1. The system is designed to monitor critical parameters such as pressure and temperature during engine operation, ensuring both safety and performance data collection. The integration approach focuses on reliability, modularity, and data accuracy while minimizing weight and power consumption.

S.No	Components	Type / Model	Quantity	Specs
1	Pressure Transducer	OMEGA PX119-1.5KGI	1	Pressure Range: 15 to 5000 psi
2	Teensy	4.1	1	Development Board
3	Thermocouple Amplifier Breakout	MAX31856	3	Universal
4	JST	2-pin JST SM Plug	1	Jst connector for battery to Teensy
5	5 V Battery	2 Cell	2	3.7V 4400mAh
6	Thermocouple	N type	3	1273 K
7	Buck Convertor	SparkFun Buck Regulator Breakout	1	32 V ~ 5 V
8	DC Current Sensor	INA219	1	26V ±3.2A Max
9	12 V Battery	3S2P	1	

Table 6.2.1 DAQ Components Lists

The data acquisition system consists of seven primary components as identified in the bill of materials:

1. Pressure Transducer (OMEGA PX119-1.5KGI): High-precision, ratio metric pressure sensor covering 15 – 5000 psi, with integrated temperature compensation and long-term stability.

2. Teensy 4.1 Development Board: 600 MHz ARM Cortex-M7 microcontroller board with abundant GPIO, SPI/I²C/UART interfaces and onboard 3.3 V regulator for your data-acquisition and control logic.

3. Thermocouple Amplifier Breakout (MAX31856) × 3: Universal thermocouple-to-SPI interface with cold-junction compensation; supports Type N (and other) thermocouples for accurate temperature readings.

4. JST Connector (2-pin JST SM Plug): Mates with matching socket to provide a secure, polarized power connection between your battery pack and the Teensy's VIN/GND.

5. 5 V Battery (2-Cell Li-Ion, 3.7 V nom, 4400 mAh) × 2: High-capacity battery cells (wired as needed) to supply 5 V rail for sensors, servo and logic (via your buck converter).

6. Thermocouple (Type N) × 3 : Sheathed high-temperature probes rated up to 1273 K for mounting on your combustion-chamber casing.

7. Buck Converter (SparkFun Buck Regulator Breakout): Step-down regulator to take your 12 V battery pack down to a stable 5 V (or VIN) for powering the Teensy rail and peripherals.

8. DC Current Sensor (INA219): I²C-interfaced high-side shunt monitor for measuring load currents and voltages on your 5 V rail (e.g. battery drain, servo draw).

9. 12 V Battery (3S2P Li-Ion Pack): Main power source (3 cells in series, 2-in-parallel) feeding the buck converter and any other 12 V loads.

6.2.1 CAD Design

The CAD design consists of three 3D-printed components that snap and screw neatly around your DAQ electronics, providing both protection and easy serviceability. The top cover is a slim, 10 mm-thick lid with precise vents along its edges for gentle airflow, a single rectangular port for your waterproof cable gland, and hidden bosses that index into the core. Below it, the red mid-section houses your main board and three daughter-boards on integrated rails and pegs—no extra brackets needed. The cutouts guide sensor wiring through the walls, while built-in clips hold each daughter-board firmly in place, and standoffs secure the top PCB with M2.5 screws.

The bottom base matches the core's 113 × 32 mm footprint and rises 33 mm to cradle pressure transducers and allow thermocouple leads to pass through internal channels. Four heat-set insert pockets align perfectly with the cores, so you can stack and fasten the parts with standard M3 screws. Shallow recesses

have been added under key standoffs to save material and improve heat transfer, and through-holes line up for silicone-grommet sealing of each sensor cable.

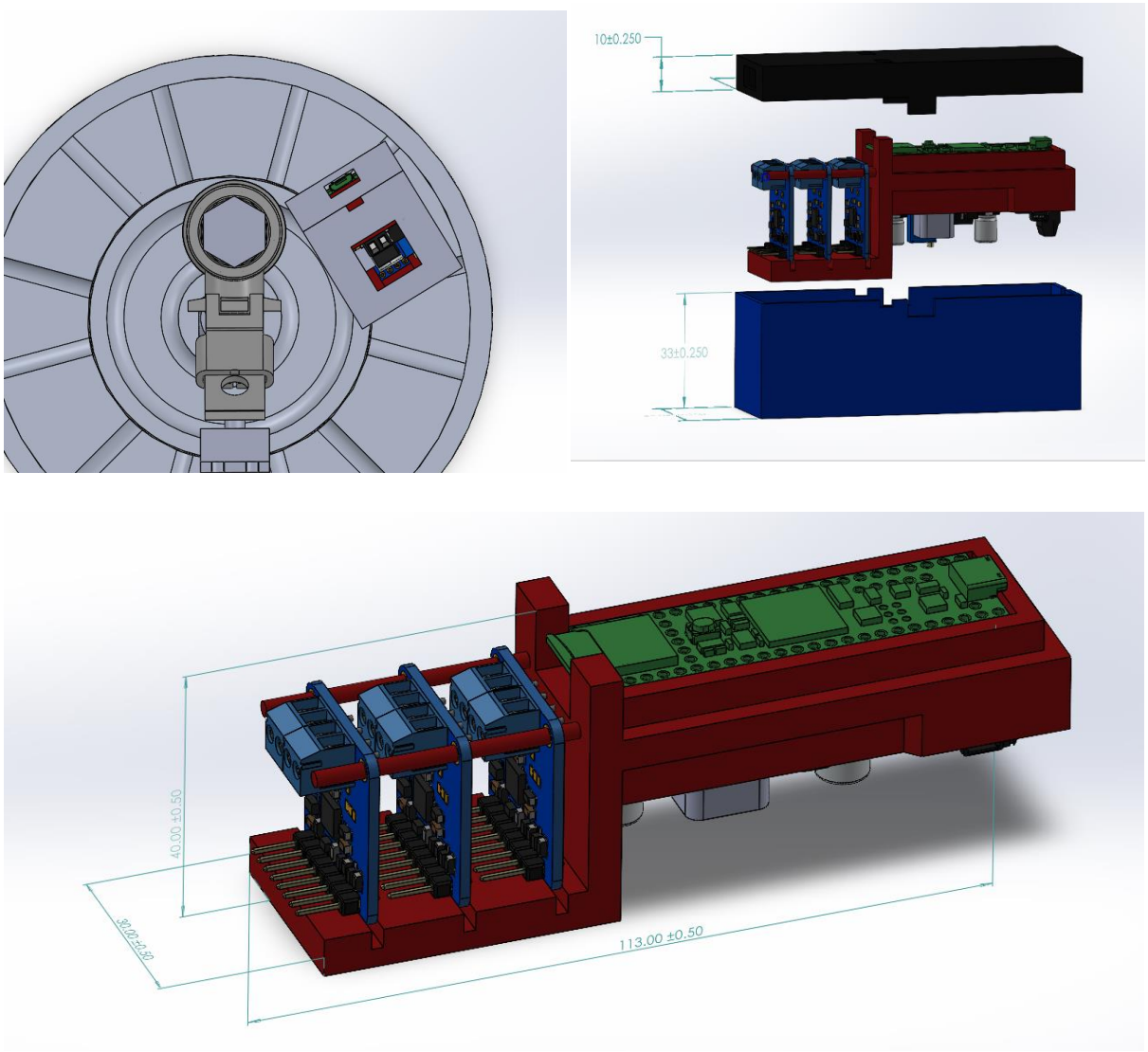


Figure 19:1 DAQ Assembly CAD

The thermocouple can be mounted to the combustor casing by crimping it. Peening or crimping deforms a small metal washer against the thermocouple junction to clamp it to the surface, providing a low-profile, minimally invasive attachment. These techniques work well at high temperatures where adhesives may fail, and they avoid introducing foreign materials that could outgas under vacuum conditions

6.2.2 Electrical Schematic

The electrical system is centred around the Teensy 4.1 microcontroller with the connections shown in Figure 6.2.1.

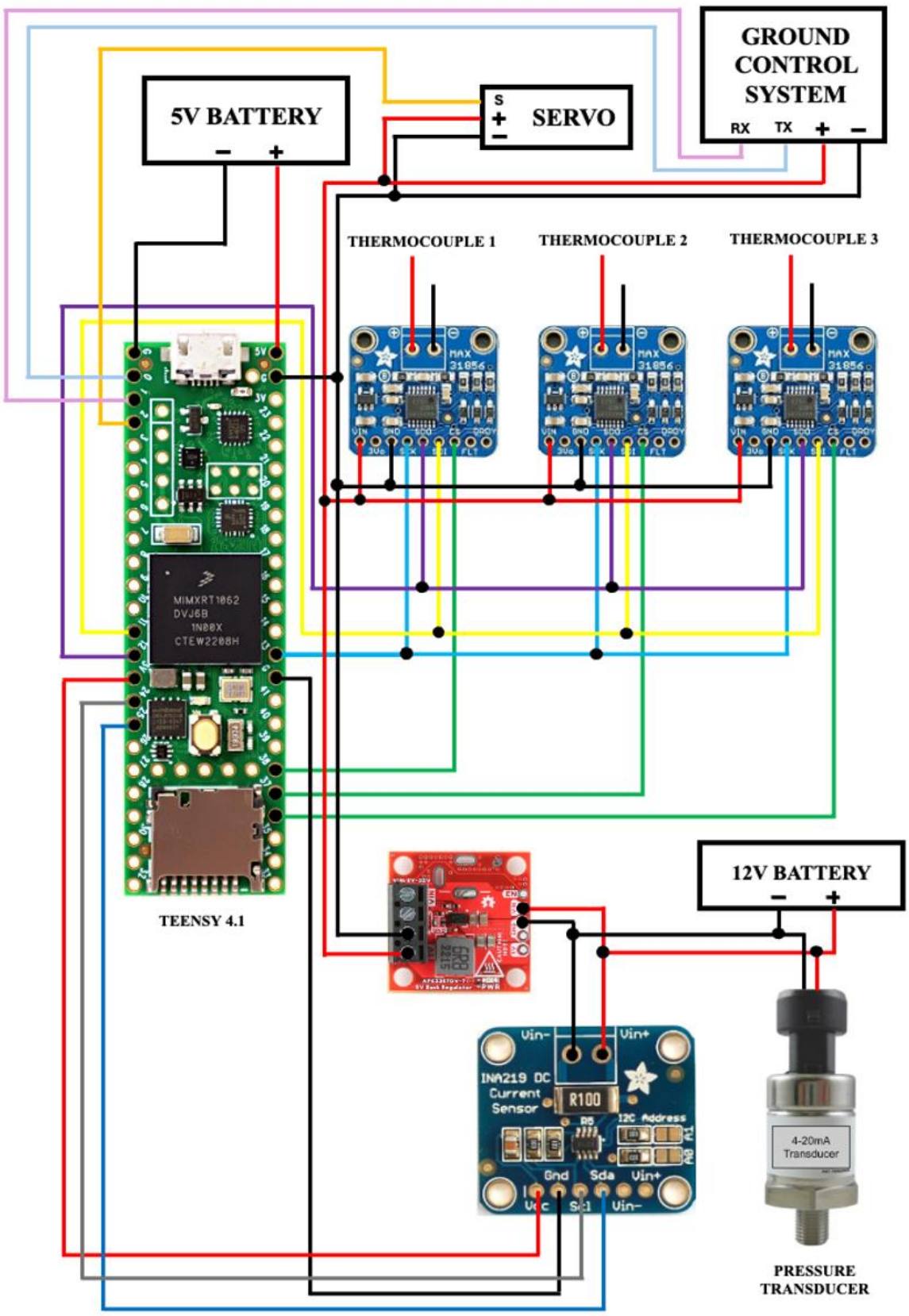


Figure 20: Electric Schematic

1. Power Supply Circuit:
    a. Battery connection to Teensy via JST connector with reverse polarity protection
    b. Step-up/step-down voltage regulation to maintain stable 5V for the Teensy
    c. Power distribution to sensors with appropriate filtering and isolation

2. Sensor Interfaces:
    a. Pressure transducer connected to Teensy analog input with signal conditioning
    b. MAX31856 breakout boards interfacing with thermocouples via SPI communication
    c. LED status indicators for power, data logging, and system status

3. Data Storage:
    a. MicroSD card connection via Teensy's onboard socket for data logging

6.2.3 Software Integration

The software architecture follows a modular design:

1. Core Firmware:
    a. Real-time operating system for precise timing and task management
    b. Sensor polling at 100Hz minimum for pressure and 10Hz for temperature
    c. Timestamped data storage with configurable sampling rates
    d. Event detection algorithms for identifying key phases (ignition, steady-state, shutdown)

2. Data Management:
    a. Redundant storage to prevent data loss
    b. Buffer management for continuous recording

3. User Interface:
    a. Pre-flight configuration via USB connection
    b. Status LEDs for critical system states
    c. Optional Bluetooth connectivity for wireless setup and monitoring

6.2.4 Integration Procedure

1. Sensor Installation:
    a. Install pressure transducer in the combustion chamber port using appropriate torque
    b. Mount thermocouples at designated locations using high-temperature adhesive and mechanical fasteners
    c. Route all sensor cables following the predetermined path
    d. Secure cables with high-temperature cable ties at 10cm intervals

2. Electronics Installation:
    a. Mount the main housing to the designated location using vibration-dampening fasteners
    b. Connect all sensor cables to the appropriate inputs
    c. Install the battery in the designated compartment
    d. Perform post-installation continuity and isolation tests

3. System Verification:
    a. Power up the system and verify status indicators
    b. Perform sensor readings verification
    c. Test data recording and retrieval
    d. Conduct simulated operation sequence

6.2.5 Testing Procedures

1. Pressure Transducer Calibration:
    a. Connect the OMEGA PX119-1.5KGI to a calibrated pressure source
    b. Apply pressures at 0%, 25%, 50%, 75%, and 100% of full scale
    c. Record output voltage at each point and verify linearity
    d. Verify temperature compensation by repeating at different ambient temperatures

2. Thermocouple System Testing:
    a. Calibrate each MAX31856 amplifier using precision temperature references
    b. Test thermocouples in an environmental chamber across the expected temperature range (-40°C to 135°C)
    c. Verify cold-junction compensation functionality
    d. Measure response time to temperature changes

3. Power System Verification:

a. Measure battery discharge curve under expected load conditions
b. Verify runtime exceeds mission duration by at least 100%
c. Test power system under vibration conditions
d. Validate battery charging system functionality

6.2.6 Operational Procedures

1. System Configuration:
    a. Program the Teensy with the latest firmware version
    b. Set sampling rates appropriate for the mission profile
    c. Format the SD card and verify storage capacity
    d. Perform full battery charge and capacity verification

2. Component Preparation:
    a. Verify Calibration of all sensors and document calibration coefficients
    b. Apply thread sealant to pressure transducer fitting
    c. Prepare thermocouple junctions with appropriate high - temperature protection
    d. Verify all connectors for proper seating and locking

3. Data Retrieval:
    a. Allow system to cool down if necessary
    b. Access the data acquisition system and power down
    c. Remove SD card or connect via USB for data download
    d. Create backup of raw data before analysis

4. System Inspection:
    a. Inspect all sensors and wiring for damage
    b. Check battery status and recharge if needed
    c. Verify housing integrity and mounting security

5. Data Analysis:
    a. Process raw data using analysis software
    b. Generate time-series plots of key parameters
    c. Calculate performance metrics (specific impulse, combustion efficiency)
    d. Compare results with theoretical predictions

This data acquisition system integrates high-quality components to provide reliable monitoring of critical parameters in a small-scale hybrid rocket engine. The OMEGA pressure transducer and MAX31856 thermocouple amplifiers provide precise measurements, while the Teensy 4.1 offers powerful processing capabilities in a compact form factor. The system design allows for both flight and ground-test configurations, with appropriate testing procedures to ensure reliability and accuracy.

By following the detailed integration, testing, and operational procedures outlined in this report, the data acquisition system will provide valuable insights into engine performance and behaviour, supporting both development efforts and operational flights. The modular approach allows for future expansion as additional measurement requirements arise.

7 Ground Testing Set-up

Several small modifications and pieces of additional equipment will be necessary to facilitate the testing of the rocket in a horizontal configuration. Firstly, the rocket will need to be supported in a way that it can be attached to the test stand at Protolaunch. For this purpose, we shall be modifying the test stand used by previous Cranseds teams by extending its length and reinforcing as necessary.

8 Component Level Tests

This section will illustrate the planned tests that will be completed ahead of the full rocket assembly. The purpose of these tests will be to validate the function of each component before any propellants are added to the system. These tests will also validate that sub-components and sub-systems work as specified by the manufacturer or as designed.

8.1 Fluid System

Component/System	Main Onboard Servo Valve
Function	Start propellant flow
Test	Pressurise complete fluid system using a pressure washer, open valve at 55 bar.
Pass Requirement	Servo successfully actuates valve. Hold pressure for 2x expected operating time without leaks or damage.

Component	Tank
Function	Hold pressurised oxidiser
Test	Pressurise with power washer (tank only).
Pass Requirement	Hold 1.5x operating pressure for 2x planned operating time.

Component	Onboard Oxidiser Lines & Fittings
Function	Hold pressurised oxidiser
Test	Pressurise with power washer as part of a total fluid system test.
Pass Requirement	Hold operating pressure for 2x planned operating time. No leaks, ruptures or deformation

Component	Ground Fill Station Nitrous Lines
Function	Feed nitrous to the vehicle
Test	Pressurise with power washer.
Pass Requirement	Hold operating pressure for 2x planned operating time. No leaks, ruptures or deformation

Component	Quick Connect
Function	Fuel the vehicle and allow for separation at launch
Test	Connect ground line to vehicle fluid system and pressurise with a power washer to 55 bar. Remotely separate lines.
Pass Requirement	Hold operating pressure for 2x expected operating time. Water must flow from one connection to another whilst connected. No leaks or deformation.

8.2 Combustion System

Component	Combustion chamber
Function	
Test	Pressurise with power washer.
Pass Requirement	Hold 1.5x operating pressure for 2x planned operating time.

9 Risk Assessment

Failure Mode	Mission Phase	Failure Probability	Mishap Severity	Criticality Ranking	Justification
Tank exploding	Ascent	1	3	Medium	Large factor of safety
Fluid Line bursting	Ascent	1	3	Medium	Large factor of safety
Bulkhead ejection	Ascent	1	3	Medium	Large factor of safety
Unresponsive flight computer	Ignition	1	2	Minor	Extensive testing of computer before flight

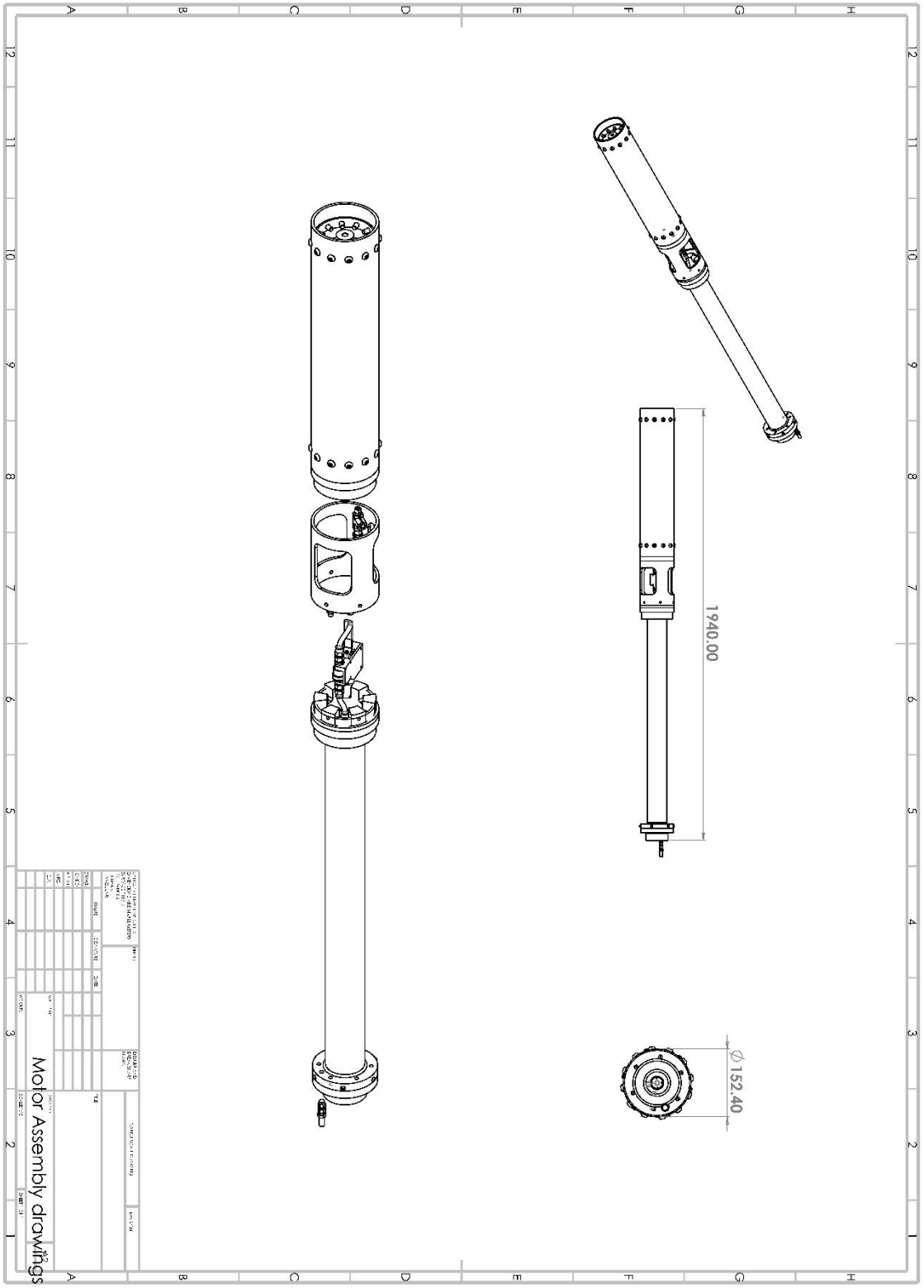
10 References

(1) Sennot A. & Sharp C. Mojave Sphinx Liquid Bipropellant Rocket: Build, Integration, and Launch Guidebook: HCR-5100(Internet). 1st ed. (Place Unknown): Self-Published; 2024 (cited 2024 November 28). Available from: https://www.halfcatrocketry.com/mojave-sphinx

(2) Sadegh A.M, marks standard handbook for mechanical engineers

11 Appendices

A1 Dimensioned CAD Drawings



A2 Mechanical Calculations

Tank Stresses

Parameters

Operating Pressure (P)	5.5 MPa	**Internal Diameter (D_i)**	139.7 mm
Wall Thickness (t)	6.35 mm	**Number of Bolts (N)**	12
Bolt Major Diameter (d_2)	10 mm	**Bolt Minor Diameter (d_1)**	8.376 mm
Edge Distance (E)	50 mm	**Safety Factor (FoS)**	3

Material Properties

Material	Property	Value
Aluminium (6061-T6)	Yield Strength (YTS)	240 MPa
	Bearing Yield Strength	386 MPa
	Shear Strength	206 MPa
Mild Steel (6.8)	Ultimate Tensile Strength	600 MPa

Design Calculation

1. Allowable Stress for Pressure Vessel

$$\sigma_{allowable} = \frac{Yield\ Tensile\ Strength\ (YTS)}{Factor\ of\ Safety\ (FoS)}$$

$$\sigma_{allowable} = \frac{240}{3} = 80\ MPa$$

2. Proof Pressure

$$P_{proof} = 2 \times P$$

$$P_{proof} = 2 \times 5.5 = 11\ MPa$$

3. Bolt Shear Stress

$$\sigma_{bolt\ shear} = \frac{\left(\pi/4\right) D_i^2\ P_{proof}}{N \cdot \left(\pi/4\right) d_1^2}$$

$$\sigma_{bolt\ shear} = \frac{\left(\pi/4\right)\ \times 139.7^2\ \times\ 11}{12 \times \left(\pi/4\right) \times\ 8.376^2} = 255\ MPa$$

$$FS_{bolt\ shear} = \frac{0.75 \times UTS}{\sigma_{bolt\ shear}}$$

$$FS_{bolt\ shear} = \frac{0.75 \times 600}{255} = 1.76$$

4. Bolt Tear-Out Stress

$$F_{bolt} = \frac{\pi}{4} D_i^2 \frac{P_{proof}}{N}$$

$$F_{bolt} = \frac{\pi}{4} \times 139.7^2\ \times \frac{11}{12} = 14050.6\ \text{MPa}$$

$$E_{min} = E - \frac{d_2}{2}$$

$$E_{min} = E - \frac{10}{2} = 45\ mm$$

$$\sigma_{tear-out} = \frac{F_{bolt}}{2 \cdot E_{min} \cdot t}$$

$$\sigma_{tear-out} = \frac{14050.6 \times 10^6}{2 \times 45 \times 6.35} = 24.6\ MPa$$

$$FS_{tear-out} = \frac{Shear\ Strength}{\sigma_{tear-out}}$$

$$FS_{tear-out} = \frac{206}{24.6} = 8.38$$

5. Casing Tensile Stress

$$\sigma_{tensile} = \frac{\left(\pi/4\right) D_i^2\ P_{proof}}{\left((D_o - t) \cdot \pi - (N \cdot d_2)\right) \cdot t}$$

$$\sigma_{tensile} = \frac{\left(\pi/4\right)\ \times 139.7^2\ \times 11}{\left((152.4 - 6.35) \cdot \pi - (12 \times 10)\right) \times 6.35} = 78.36\ MPa$$

$$FS_{tensile} = \frac{YTS}{\sigma_{tensile}}$$

$$FS_{tensile} = \frac{240}{78.36} = 3.06$$

6. Bearing Stress

$$\sigma_{bearing} = \frac{F_{bolt}}{d_2 \cdot t}$$

$$\sigma_{bearing} = \frac{14050.6 \times 10^6}{10 \times 6.35} = 221.3\ MPa$$

$$FS_{bearing} = \frac{YTS}{\sigma_{bearing}}$$

$$FS_{bearing} = \frac{240}{22.13} = 1.74$$

Combustion Chamber Stresses

Parameters

Operating Pressure (P)	2.11 MPa	**Internal Diameter (D_i)**	82.5 mm
Wall Thickness (t)	3.2 mm	**Number of Bolts (N)**	6
Bolt Major Diameter (d_2)	10 mm	**Bolt Minor Diameter (d_1)**	8.376 mm
Edge Distance (E)	30 mm	**Safety Factor (FoS)**	3

Material Properties

Material	Property	Value
Aluminium (6061-T6)	Yield Strength (YTS)	240 MPa
	Bearing Yield Strength	386 MPa
	Shear Strength	206 MPa
Mild Steel (6.8)	Ultimate Tensile Strength	600 MPa

Design Calculation

1. Allowable Stress for Pressure Vessel

$$\sigma_{allowable} = \frac{Yield\ Tensile\ Strength\ (YTS)}{Factor\ of\ Safety\ (FoS)}$$

$$\sigma_{allowable} = \frac{240}{3} = 80\ MPa$$

2. Proof Pressure

$$P_{proof} = 2 \times P$$

$$P_{proof} = 2 \times 3 = 6\ MPa$$

3. Bolt Shear Stress

$$\sigma_{bolt\ shear} = \frac{\left(\pi/4\right) D_i^2\ P_{proof}}{N \cdot \left(\pi/4\right) d_1^2}$$

$$\sigma_{bolt\ shear} = \frac{\left(\pi/4\right) \times 82.5^2\ \times\ 6}{6 \times \left(\pi/4\right) \times\ 8.376^2} = 97\ MPa$$

$$FS_{bolt\ shear} = \frac{0.75 \times UTS}{\sigma_{bolt\ shear}}$$

$$FS_{bolt\ shear} = \frac{0.75 \times 600}{97} = 4.64$$

4. Bolt Tear-Out Stress

$$F_{bolt} = \frac{\pi}{4} D_i^2 \frac{P_{proof}}{N}$$

$$F_{bolt} = \frac{\pi}{4} \times 82.5^2\ \times \frac{11}{6} = 5345.6\ \text{MPa}$$

$$E_{min} = E - \frac{d_2}{2}$$

$$E_{min} = E - \frac{10}{2} = 25\ mm$$

$$\sigma_{tear-out} = \frac{F_{bolt}}{2 \cdot E_{min} \cdot t}$$

$$\sigma_{tear-out} = \frac{1405 \times 10^6}{2 \times 45 \times 6.35} = 33.4\ MPa$$

$$FS_{tear-out} = \frac{Shear\ Strength}{\sigma_{tear-out}}$$

$$FS_{tear-out} = \frac{206}{24.6} = 6.17$$

5. Casing Tensile Stress

$$\sigma_{tensile} = \frac{\left(\pi/4\right) D_i^2 \, P_{proof}}{\left((D_o - t) \cdot \pi - (N \cdot d_2)\right) \cdot t}$$

$$\sigma_{tensile} = \frac{\left(\pi/4\right) \times 82.5^2 \times 6}{\left((88.9 - 3.2) \cdot \pi - (6 \times 10)\right) \times 3.2} = 48\ MPa$$

$$FS_{tensile} = \frac{YTS}{\sigma_{tensile}}$$

$$FS_{tensile} = \frac{240}{78.36} = 5.01$$

6. Bearing Stress

$$\sigma_{bearing} = \frac{F_{bolt}}{d_2 \cdot t}$$

$$\sigma_{bearing} = \frac{5345.6 \times 10^6}{10 \times 3.2} = 167\ MPa$$

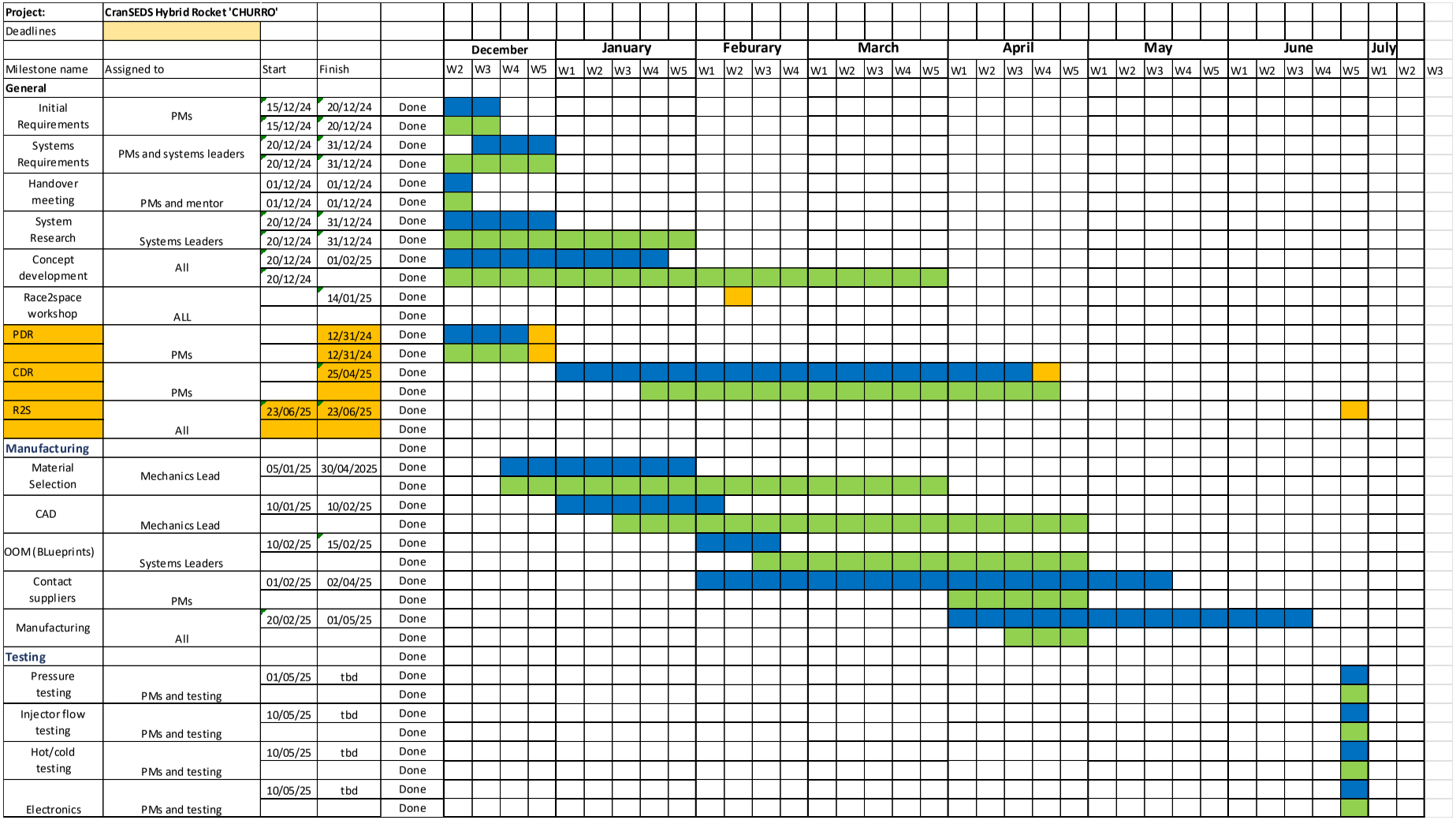
$$FS_{bearing} = \frac{YTS}{\sigma_{bearing}}$$

$$FS_{bearing} = \frac{240}{22.13} = 2.31$$

A3 Bill Of Materials

Price	#	Cost	Item	Supplier	Item #	Part	URL	Note			
32	1	32	Aluminium pipe				https://www.aluminiumwarehouse.co.uk/products/3-1-2-in-x-10-swg-aluminium-round-tube			TOTAL	1931.56
100	1	100	Aluminium solid rod				https://www.aluminiumwarehouse.co.uk/collections/aluminium-round-bar/products/3-1-4-in-diameter-aluminium-round-bar				
60.55	1	60.55	Aluminium solid rod				https://www.aluminiumwarehouse.co.uk/collections/aluminium-round-bar/products/6-in-diameter-aluminium-round-bar-2011				
10	1	10	Aluminium plate				5mm - 6082 - Aluminium Plate				
600	1	600	Phenolic rod				https://www.theplasticshop.co.uk/tufnol-rod-whale-60mm-70mm-80mm-90mm-100mm-diameter.html				
36	1	36	Phenolic pipe				https://modelrockets.co.uk/scratch-building-bits/body-tubes-air-frames/76mm-x-1000mm-seamless-phenolic-airframe-tube/?utm_source=chatgpt.com				
43.22	1	43.22	HDPE rod				HDPE Rod | Direct Plastics Limited				
23.77	1	23.77	HDPE rod				https://www.directplastics.co.uk/hdpe-rod?gad_source=1&gclid=Cj0KCQjw782_BhDjARIsABTv_JA7Hgt17hzU7T_-8k7L8xip0ie4Fsk1hTu34hiuIqy2fy2qB4ZQorAaAsthEALw_wcB				
24	1	24	Rubber Sheet				https://www.amazon.co.uk/Neoprene-Solid-Rubber-Sheet-1-5mm/dp/B00FF5RSB4?				
24.49	2	48.98	4.1 Development Board			Teensy	https://www.adafruit.com/product/4622				
17.5	3	52.5	MAX31856			Thermocouple Amplifier Breakout	https://www.adafruit.com/product/3263				
0.58	4	2.32	2-pin JST SM Plug			JST	https://www.adafruit.com/product/2880				
15.4	2	30.8	2 Cell			Lithium Ion Battery	https://www.adafruit.com/product/354				
5.36	1	5.36	LiPo/LiIon Batt w/MicroUSB Jack - v1			Battery Charger	https://www.adafruit.com/product/1904				
47.08	1	47.08				Arduino Uno	https://store.arduino.cc/products/arduino-uno-wifi-rev2				
6.5	1	6.5				USB Cable	https://store.arduino.cc/products/usb-2-0-cable-type-a-b?_pos=1&_psq=M000006+OR+C000141+OR+TPX00018&_ss=e&_v=1.0				
7.49	1	7.49	64GB SD Card			SD Card	https://www.amazon.co.uk/SanDisk-Ultra-microSDXC-140MB-Adapter/dp/B0B7NXBM6P				
4.39	1	4.39				Jumper Cables	https://www.amazon.co.uk/120pcs-Multicolored-Breadboard-Compatible-Arduino/dp/B0DP1YNH7H/				
5.5	1	5.5				Breadboard	https://store.arduino.cc/products/breadboard-840-contacts				
3.3	1	3.3				Buck Converter	https://thepihut.com/products/dc-dc-buck-mini-module-3-3v-5v				
43.04	1	43.04				Thermocouple	https://uk.rs-online.com/web/p/thermocouples/2740184	N-Type			
7.96	1	7.96				Thermocouple Fitting	https://uk.rs-online.com/web/p/sensor-accessories/3817277				
23.1	1	23.1				Thermocouple Amplifier Breakout	https://shop.pimoroni.com/products/mcp9600-thermocouple-amplifier-breakout?variant=21439215272019				
90	1	90				Pressure Transducer	https://strainsense.store/product/industrial-pressure-transmitter-0-5-vdc-gaugeabsolutecompound/				
1.45	1	1.45				Flare Nut	https://www.bes.co.uk/oil-flare-nut-6mm-15848/				
2.08	1	2.08				Male Flare Adapter					
14.99	1	14.99				Battery	https://www.amazon.co.uk/Rechargeable-Lithium-Batteries-Capacity-Charging/dp/B0CKN34L6K				
0	1	0				Oxygen Tank					
3	1	3	Nylon Pipe			6mm x 3mm Plastic Nylon Tube Semi-rigid PA 12 HL DIN 73378	https://www.advancedfluidsolutions.co.uk/6mm-x-3mm-plastic-nylon-tube-semi-rigid-pa-12-hl-din-73378-1967-p.asp				
0	1	0				Electronic Valve (Josh wants to use same as oxidisor valve)					
10	1	10				High Voltage Transform Generator	https://www.amazon.co.uk/Shinesline-Voltage-Transformer-Generator-400000V/dp/B01MXV64IY/				
6	1	6				Wires	https://www.amazon.co.uk/QWORK-Meter-22AWG-Electrical-Extension/dp/B0BG54KWG2/				
12	1	12				Tube to tank connector	https://www.advancedfluidsolutions.co.uk/6mm-x-18-bspt-80-bar-hp-brass-metal-push-fit-straight-stud-fitting-1965-p.asp				
28.99	1	28.99	25 kg High Torque Metal Geared Digital Brushless S			Servo-Motor	https://howesmodels.co.uk/product/25kg-high-torque-metal-geared-waterproof-digital-brushless-servo-bl-s3225/				
31.44	4	125.76	AN6 PTFE Hose Fitting for Braided AN PTFE Lines			Hose Fitting	https://auto.helperformance.com/an6-straight-ptfe-hose-fitting-for-braided-an-ptfe-lines				
13.64	1	13.64	AN6 PTFE Stainless Braided Fuel Oil Hose Line			Braided Hose	https://auto.helperformance.com/hel-stainless-steel-6-an-braided-ptfe-hose-8mm-internal-diameter				
26.96	4	107.84	AN6 to 1/4" BSPP Male to Male Adaptors			AN6 to BSPP adapters	https://www.ebay.co.uk/itm/360632843339?chn=ps&_ul=GB&google_free_listing_action=view_item&gQT=0				
22.26	1	22.26	1/4" Stainless Steel 2-way Ball Valve PTFE Seals Rat			Main Valve	https://tameson.co.uk/products/bl2sm-014-g1-4inch-2-way-ball-valve-stainless-steel-ptfe-63-bar				
25.66	1	25.66	Thread Sealing Tape High Density PTFE - rated up to			PTFE Tape	https://tameson.co.uk/products/thread-sealing-tape-high-density-ptfe				
7.08	2	14.16	AN4 PTFE Hose Fittings for Braided PTFE Line			Hose Fitting	https://www.speedingparts.co.uk/p/fuelsystem-an-fittings/an-fittings-an-hose/spd-an-fittings/hose-ends-ptfe/straight-ptfe-hose-ends/an4-hose-end-straight-ptfe.html				
10.8	2	21.6	AN4 to 1/8" BSPP Mate to Male			AN4 to BSPP	https://www.speedingparts.co.uk/p/fuelsystem-an-fittings/an-fittings-an-hose/spd-an-fittings/an-pipe-thread-adapter/an4-1828-pipe-thread-adapter.html				
8.01	1	8.01	AN4 PTFE Stainless Braided Hose			Braided Hose	https://www.speedingparts.co.uk/p/fuelsystem-an-fittings/an-fittings-an-hose/an-hose/steel-braided-ptfe-an-hose/an4-steel-braided-hose-ptfe.html?gQT=1				
17.4	1	17.4	Hydraulic Quick Disconnect - Female			Quick Disconnect - F	https://uk.rs-online.com/web/p/hydraulic-quick-connect-couplings/7335615?gb=s				
8.16	1	8.16	Hydraulic Quick Disconnect - Male			Quick Disconnect - M	https://uk.rs-online.com/web/p/hydraulic-quick-connect-couplings/7335612?gb=s				
175.65	1	175.65	Stainless Steel High Pressure Proportional Relief Va			Relief Valve	https://products.swagelok.com/en/c/high-pressure-relief-valve/p/SS-4R3A5				
5.05	1	5.05	Purple Spring Kit for R3A Series Proportional Relief			Spring Kit (Flight Pressure)	https://products.swagelok.com/en/c/spring-kit/p/177-R3A-K1-C				
		0									

A4 Gantt Chart



Project:	CranSEDS Hybrid Rocket 'CHURRO'			
Deadlines				
Milestone name	Assigned to	Start	Finish	
General				
Initial Requirements	PMs	15/12/24	20/12/24	Done
		15/12/24	20/12/24	Done
Systems Requirements	PMs and systems leaders	20/12/24	31/12/24	Done
		20/12/24	31/12/24	Done
Handover meeting	PMs and mentor	01/12/24	01/12/24	Done
		01/12/24	01/12/24	Done
System Research	Systems Leaders	20/12/24	31/12/24	Done
		20/12/24	31/12/24	Done
Concept development	All	20/12/24	01/02/25	Done
		20/12/24		Done
Race2space workshop	ALL		14/01/25	Done
				Done
PDR	PMs		12/31/24	Done
			12/31/24	Done
CDR	PMs		25/04/25	Done
				Done
R2S	All	23/06/25	23/06/25	Done
				Done
Manufacturing				Done
Material Selection	Mechanics Lead	05/01/25	30/04/2025	Done
				Done
CAD	Mechanics Lead	10/01/25	10/02/25	Done
				Done
OOM (BLueprints)	Systems Leaders	10/02/25	15/02/25	Done
				Done
Contact suppliers	PMs	01/02/25	02/04/25	Done
				Done
Manufacturing	All	20/02/25	01/05/25	Done
				Done
Testing				Done
Pressure testing	PMs and testing	01/05/25	tbd	Done
				Done
Injector flow testing	PMs and testing	10/05/25	tbd	Done
				Done
Hot/cold testing	PMs and testing	10/05/25	tbd	Done
				Done
Electronics	PMs and testing	10/05/25	tbd	Done
				Done

A5 Combustion chamber Heat Transfer Matlab Code

```
% Transient Heat Conduction in a Two-Layer Hollow Cylinder (FTCS Scheme)
% Convection at inner surface, constant temperature at outer surface

clear; clc; close all;

%% --- Parameters ---

% Geometry
r_in = 0.02;          % Inner radius (m)
r_layer1 = 0.04;      % Outer radius of insulation (m)
r_layer2 = 0.06;      % Outer radius of metal casing (m)
r_out = r_layer2;

% Materials
k1 = 0.01; k2 = 237;
rho1 = 10; rho2 = 2699;
cp1 = 1795; cp2 = 900;

alpha1 = k1 / (rho1 * cp1);
alpha2 = k2 / (rho2 * cp2);

% Boundary Conditions
h_in = 430;           % Convective heat transfer coefficient (W/m².K)
T_fluid = 2000;       % Inner fluid temperature (K)
T_outer = 300;        % Fixed outer wall temperature (K)
T_init = 300;         % Initial temperature (K)

% Simulation Controls
nr = 30;                        % Number of radial nodes
r = linspace(r_in, r_out, nr)'; % Radial grid
dr = (r_out - r_in) / (nr - 1); % Radial step size

t_total = 20;       % Total time for simulation (s)
alpha_max = max([alpha1 alpha2]);
max_dt = 0.5 * dr^2 / alpha_max;
dt = min(0.01, max_dt); % Stable time step
nt = round(t_total / dt);

%% --- Setup ---

% Initial temperature distribution
T = ones(nr, 1) * T_init;

% Assign properties based on layer
idx1 = find(r <= r_layer1);
idx2 = find(r > r_layer1);

k = zeros(nr, 1); alpha = zeros(nr, 1);
k(idx1) = k1; alpha(idx1) = alpha1;
k(idx2) = k2; alpha(idx2) = alpha2;

% Radial location to monitor
r_target = 0.04; % You can change this value
[~, idx_target] = min(abs(r - r_target));
```

```
T_target_history = zeros(nt, 1);

%% --- Time-stepping (FTCS) ---

for t_step = 1:nt
    T_new = T;

    % Inner surface (r = r_in): Convection
    T_new(1) = (T(2) + (h_in * dr / k1) * T_fluid) / (1 + h_in * dr / k1);

    % Interior points
    for i = 2:nr-1
        T_new(i) = T(i) + alpha(i) * dt * ...
            (T(i+1) - 2*T(i) + T(i-1)) / dr^2;
    end

    % Outer surface (r = r_out): Constant Temperature
    T_new(nr) = T_outer;

    % Update temperature
    T = T_new;

    % Store temperature at target location
    T_target_history(t_step) = T(idx_target);
end

%% --- Plotting ---

time = linspace(0, t_total, nt);

figure;
plot(time, T_target_history, 'b-', 'LineWidth', 2);
xlabel('Time (s)');
ylabel(['Temperature at r = ', num2str(r_target), ' m (K)']);
title(['Temperature vs Time at r = ', num2str(r_target), ' m']);
grid on;
```

A6 MATLAB code for Nozzle geometry

This MATLAB code is designed to create and visualize the two-dimensional geometry of two types of rocket nozzles: a conical nozzle and a bell-shaped (contoured) nozzle. Rocket nozzles are critical components in rocket engines, as they accelerate exhaust gases to produce thrust. The code combines thermodynamic principles and geometric definitions to calculate the dimensions and shapes of the nozzles based on user-defined parameters.

The code starts by defining the inputs: pressure and temperature inside the combustion chamber, ambient pressure at the nozzle exit, exhaust gas molar mass, specific heat ratio, specific gas constant, specific heat at constant pressure, desired thrust, divergence angle for the conical nozzle, tangent of the divergence angle.

The code performs calculations based on isentropic flow theory, which assumes an ideal, frictionless flow without heat losses. The goal is to determine the key dimensions of the nozzle, such as the areas at the throat and the exit, which are critical for achieving the desired thrust. The first step is to calculate the expansion ratio, which represents how much the gas expands from the throat to the exit. This ratio is determined using an isentropic relation that depends on the chamber and exit pressures, as well as the specific heat ratio of the gas:

$$\varepsilon = \frac{\left(\frac{2}{\gamma+1}\right)^{\frac{1}{\gamma-1}} \cdot \left(\frac{p_c}{p_s}\right)^{\frac{1}{\gamma}}}{\sqrt{\frac{\gamma+1}{\gamma-1} \cdot \left(1 - \left(\frac{p_s}{p_c}\right)^{\frac{\gamma-1}{\gamma}}\right)}}$$

Next, the code computes the thrust coefficient, a dimensionless factor that relates the actual thrust to the theoretical thrust based on the chamber pressure and throat area. Using the thrust coefficient and the chamber pressure, the code then calculates the throat area, which is the smallest cross-sectional area where the flow reaches the speed of sound.

$$C_F = \sqrt{\frac{2\gamma^2}{\gamma-1} \cdot \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{\gamma-1}} \cdot \left(1 - \left(\frac{p_s}{p_c}\right)^{\frac{\gamma-1}{\gamma}}\right)} \quad ; \quad A_t = \frac{Thrust}{C_F \cdot p_c}$$

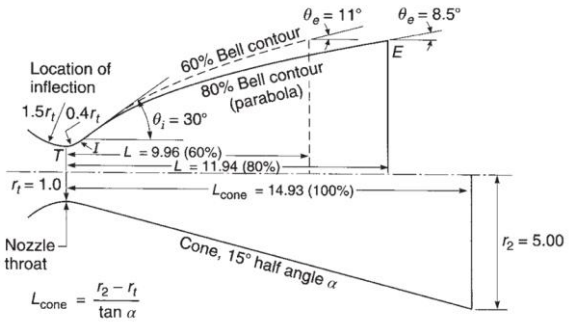
With the throat area determined, the code calculates the exit area by multiplying the throat area by the expansion ratio. Since the nozzle is assumed to have circular cross-sections, the throat and exit radii are derived from their respective areas using the formula for the area of a circle.

$$A_e = \varepsilon \cdot A_t \quad ; \quad r_t = \sqrt{\frac{A_t}{\pi}} \quad ; \quad r_e = \sqrt{\frac{A_e}{\pi}}$$

The code then calculates the length of the conical nozzle, which depends on the throat and exit radii, as well as the divergence angle. This length represents the axial distance from the throat to the exit for the conical design. Finally, the length of the bell-shaped nozzle is set as a fraction of the conical nozzle length, reflecting the more compact and efficient design of a contoured nozzle.

$$L_{cone} = \frac{r_t \cdot \left(\sqrt{\varepsilon} - 1\right) + 1.5 \cdot r_t \cdot (\sec(\alpha) - 1)}{\tan(\alpha)}$$

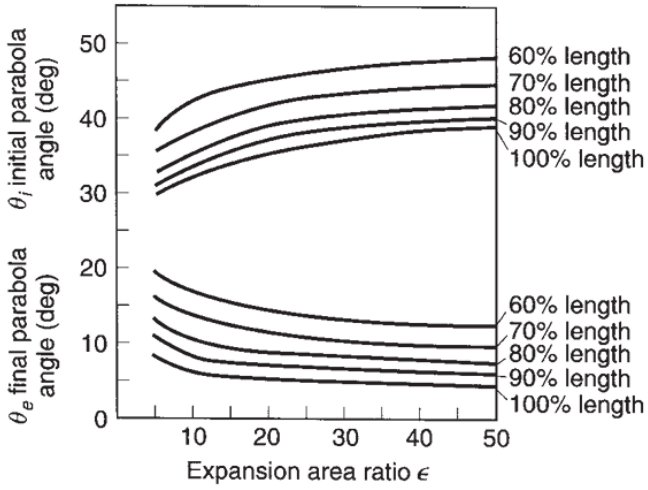
The design of the belled shape nozzle is based on the design recommendations stablished by George P.Sutton in his book "Rocket propulsion elements" as shown in the following image:



Finally, to get the geometric profiles of both the conical and bell-shaped nozzles. The conical nozzle is straightforward, while the bell-shaped nozzle requires a more complex approach to achieve its contoured shape.

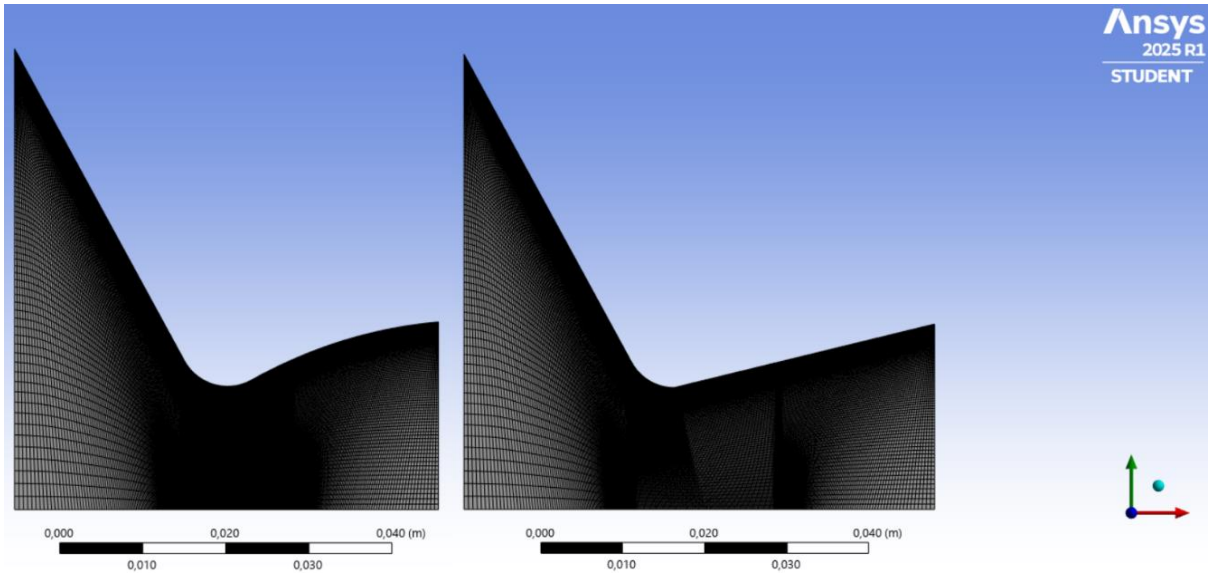
For the conical nozzle, the code defines a straight line from the throat to the exit. The throat is positioned at the origin of the coordinate system, and the exit is located at the calculated length of the conical nozzle, with the radius expanding linearly according to the divergence angle. This creates a simple, cone-like shape that is easy to manufacture but less efficient in terms of flow performance.

The bell-shaped nozzle, on the other hand, is designed to optimize the flow by minimizing losses due to flow separation. Its shape is defined in two parts. First, near the throat, the code creates a circular arc that starts at the throat and ends at an inflection point. This arc ensures a smooth transition from the throat to the diverging section. The arc is defined by a radius proportional to the throat radius and spans a specific angular range, with its centre positioned slightly above the throat. The second part of the bell-shaped nozzle is a cubic curve that extends from the inflection point to the exit. To create this curve, the code defines angles at the inflection point and the exit, which control the slope of the contour at these locations. It then sets up a system of equations to fit a cubic polynomial that passes through the inflection and exit points while matching the specified slopes. The angles of the inlet and outlet points for the parabola are normally defined according to the following graph:



A7 Nozzle CFD simulations set up

Finally,



A8 EUROC Requirements

As our motor was initially designed to be integrated into CranSED's European Rocketry Challenge (EuRoC) rocket, we have matched requirements for this competition as well.

KEY			
CRAN-R2S-25-XXX-##			
XXX -	Location on Rocket/ Requirement Type		
## -	Code		

Rocket Location	CODE	REF CODE	Note	EuRoC RQT
Nozzle (PropSys)	CRAN-R2S-25-NOZ	01	General Propulsion Systems Standards	
Nozzle (PropSys)	CRAN-R2S-25-NOZ	02	Non-toxic fuel	EuRoC-LV-RQT-0010
Nozzle (PropSys)	CRAN-R2S-25-NOZ	03	Electronics redundancy & wiring	EuRoC-LV-RQT-0020
Nozzle (PropSys)	CRAN-R2S-25-NOZ	04	15m dist. Arm/disarm ignition circuits	EuRoC-LV-RQT-0030
Nozzle (PropSys)	CRAN-R2S-25-NOZ	05	Ensure lift-off only with min.threshold force	EuRoC-LV-RQT-0040
Nozzle (PropSys)	CRAN-R2S-25-NOZ	06	Flight stability above min. threshold force	EuRoC-LV-RQT-0050
Nozzle (PropSys)	CRAN-R2S-25-NOZ	07	Launch detection arms 2ndary prop. sys.	EuRoC-LV-RQT-0060
Nozzle (PropSys)	CRAN-R2S-25-NOZ	08	Launch detection arms upper stage prop. sys.	EuRoC-LV-RQT-0070
Nozzle (PropSys)	CRAN-R2S-25-NOZ	09	EuRoC list of COTS solid motors	EuRoC-LV-RQT-0080
Nozzle (PropSys)	CRAN-R2S-25-NOZ	10	EuRoC solid motor electronic ignition system	EuRoC-LV-RQT-0090
Nozzle (PropSys)	CRAN-R2S-25-NOZ	11	Externally pressurize gaseous/liquid propellant systems.	EuRoC-LV-RQT-0100
Nozzle (PropSys)	CRAN-R2S-25-NOZ	12	Pressurize tanks after disconnecting loading lines.	EuRoC-LV-RQT-0110
Nozzle (PropSys)	CRAN-R2S-25-NOZ	13	Use dissimilar connectors for loading lines.	EuRoC-LV-RQT-0120
Nozzle (PropSys)	CRAN-R2S-25-NOZ	14	Emergency release mechanism for remote-controlled loading.	EuRoC-LV-RQT-0130
Nozzle (PropSys)	CRAN-R2S-25-NOZ	15	Accessible fluid connections from ground.	EuRoC-LV-RQT-0140
Nozzle (PropSys)	CRAN-R2S-25-NOZ	16	Demonstrate liquid fuel	EuRoC-LV-RQT-0150

			loading/unloading readiness.	
Nozzle (PropSys)	CRAN-R2S-25-NOZ	17	Facilitate oxidizer tank venting.	EuRoC-LV-RQT-0160
Nozzle (PropSys)	CRAN-R2S-25-NOZ	18	Passive PRD in isolated pressurized sections.	EuRoC-LV-RQT-0170
Nozzle (PropSys)	CRAN-R2S-25-NOZ	19	PRD discharge coefficient equal/higher than other interfaces.	EuRoC-LV-RQT-0180
Nozzle (PropSys)	CRAN-R2S-25-NOZ	20	Remote-controlled venting/offloading for launch abort.	EuRoC-LV-RQT-0190
Nozzle (PropSys)	CRAN-R2S-25-NOZ	21	Pressure test combustion chambers.	EuRoC-LV-RQT-0200
Nozzle (PropSys)	CRAN-R2S-25-NOZ	22	Leak-proof combustion chambers for testing.	EuRoC-LV-RQT-0210
Nozzle (PropSys)	CRAN-R2S-25-NOZ	23	Complete propellant loading/offloading test.	EuRoC-LV-RQT-0220
Nozzle (PropSys)	CRAN-R2S-25-NOZ	24	Complete static hot-fire test.	EuRoC-LV-RQT-0230
Combustion Chamber	CRAN-R2S-25-COM	01	Non-toxic fuel	EuRoC-LV-RQT-0010
Combustion Chamber	CRAN-R2S-25-COM	02	Electronics redundancy & wiring	EuRoC-LV-RQT-0020
Combustion Chamber	CRAN-R2S-25-COM	03	15m dist. Arm/disarm ignition circuits	EuRoC-LV-RQT-0030
Combustion Chamber	CRAN-R2S-25-COM	04	Ensure lift-off only with min. threshold force	EuRoC-LV-RQT-0040
Combustion Chamber	CRAN-R2S-25-COM	05	Flight stability above min. threshold force	EuRoC-LV-RQT-0050
Combustion Chamber	CRAN-R2S-25-COM	06	Launch detection arms 2ndary prop. sys.	EuRoC-LV-RQT-0060
Combustion Chamber	CRAN-R2S-25-COM	07	Launch detection arms upper stage prop. sys.	EuRoC-LV-RQT-0070
Combustion Chamber	CRAN-R2S-25-COM	08	EuRoC list of COTS solid motors	EuRoC-LV-RQT-0080

Combustion Chamber	CRAN-R2S-25-COM	09	EuRoC solid motor electronic ignition system	EuRoC-LV-RQT-0090
Combustion Chamber	CRAN-R2S-25-COM	10	Complete static hot-fire test.	EuRoC-LV-RQT-0230
Fuel System	CRAN-R2S-25-FSY	01	Non-toxic fuel	EuRoC-LV-RQT-0010
Fuel System	CRAN-R2S-25-FSY	02	Externally pressurize gaseous/liquid propellant systems.	EuRoC-LV-RQT-0100
Fuel System	CRAN-R2S-25-FSY	03	Pressurize tanks after disconnecting loading lines.	EuRoC-LV-RQT-0110
Fuel System	CRAN-R2S-25-FSY	04	Use dissimilar connectors for loading lines.	EuRoC-LV-RQT-0120
Fuel System	CRAN-R2S-25-FSY	05	Emergency release mechanism for remote-controlled loading.	EuRoC-LV-RQT-0130
Fuel System	CRAN-R2S-25-FSY	06	Accessible fluid connections from ground.	EuRoC-LV-RQT-0140
Fuel System	CRAN-R2S-25-FSY	07	Demonstrate liquid fuel loading/unloading readiness.	EuRoC-LV-RQT-0150
Fuel System	CRAN-R2S-25-FSY	08	Facilitate oxidizer tank venting.	EuRoC-LV-RQT-0160
Fuel System	CRAN-R2S-25-FSY	09	Passive PRD in isolated pressurized sections.	EuRoC-LV-RQT-0170
Fuel System	CRAN-R2S-25-FSY	10	PRD discharge coefficient equal/higher than other interfaces.	EuRoC-LV-RQT-0180
Fuel System	CRAN-R2S-25-FSY	11	Remote-controlled venting/offloading for launch abort.	EuRoC-LV-RQT-0190
NO2 Tank	CRAN-R2S-25-NOT	01	Non-toxic fuel	EuRoC-LV-RQT-0010
NO2 Tank	CRAN-R2S-25-NOT	02	Externally pressurize gaseous/liquid propellant systems.	EuRoC-LV-RQT-0100

NO2 Tank	CRAN-R2S-25-NOT	03	Pressurize tanks after disconnecting loading lines.	EuRoC-LV-RQT-0110
NO2 Tank	CRAN-R2S-25-NOT	04	Use dissimilar connectors for loading lines.	EuRoC-LV-RQT-0120
NO2 Tank	CRAN-R2S-25-NOT	05	Emergency release mechanism for remote-controlled loading.	EuRoC-LV-RQT-0130
NO2 Tank	CRAN-R2S-25-NOT	06	Accessible fluid connections from ground.	EuRoC-LV-RQT-0140
NO2 Tank	CRAN-R2S-25-NOT	07	Demonstrate liquid fuel loading/unloading readiness.	EuRoC-LV-RQT-0150
NO2 Tank	CRAN-R2S-25-NOT	08	Facilitate oxidizer tank venting.	EuRoC-LV-RQT-0160
NO2 Tank	CRAN-R2S-25-NOT	09	Passive PRD in isolated pressurized sections.	EuRoC-LV-RQT-0170
NO2 Tank	CRAN-R2S-25-NOT	10	PRD discharge coefficient equal/higher than other interfaces.	EuRoC-LV-RQT-0180
NO2 Tank	CRAN-R2S-25-NOT	11	Remote-controlled venting/offloading for launch abort.	EuRoC-LV-RQT-0190
NO2 Tank	CRAN-R2S-25-NOT	12	Complete propellant loading/offloading test.	EuRoC-LV-RQT-0220
NO2 Tank	CRAN-R2S-25-NOT	13	SRAD press. vess. must use burst disc 6mm min dia.	EuRoC-LV-RQT-0560
NO2 Tank	CRAN-R2S-25-NOT	14	Burst discs rupture at ≤1.25 times nominal tank pressure.	EuRoC-LV-RQT-0570
NO2 Tank	CRAN-R2S-25-NOT	15	Mark burst discs with rupture pressure and identifier.	EuRoC-LV-RQT-0580
NO2 Tank	CRAN-R2S-25-NOT	16	Use same stock material for burst discs.	EuRoC-LV-RQT-0590

NO2 Tank	CRAN-R2S-25-NOT	17	Additional relief device in pressure vessels.	EuRoC-LV-RQT-0600
NO2 Tank	CRAN-R2S-25-NOT	18	MPV designed to burst at ≥2 times max pressure.	EuRoC-LV-RQT-0610
NO2 Tank	CRAN-R2S-25-NOT	19	CPV designed to burst at ≥3 times max pressure.	EuRoC-LV-RQT-0620
NO2 Tank	CRAN-R2S-25-NOT	20	Test and document burst discs.	EuRoC-LV-RQT-0630
NO2 Tank	CRAN-R2S-25-NOT	21	Proof pressure test pressure vessels at 1.5x max press.	EuRoC-LV-RQT-0640
Parachute	CRAN-R2S-25-PAR	01	Implement redundant recovery system electronics.	EuRoC-LV-RQT-0240
Parachute	CRAN-R2S-25-PAR	02	Use COTS flight computer for redundant recovery.	EuRoC-LV-RQT-0250
Parachute	CRAN-R2S-25-PAR	03	Accessible electronics switches/connectors.	EuRoC-LV-RQT-0260
Parachute	CRAN-R2S-25-PAR	04	Electronics accessible from ground.	EuRoC-LV-RQT-0270
Parachute	CRAN-R2S-25-PAR	05	Mount electronics on side opposite to launch rail.	EuRoC-LV-RQT-0280
Parachute	CRAN-R2S-25-PAR	06	Comply with energetic device requirements.	EuRoC-LV-RQT-0290
Parachute	CRAN-R2S-25-PAR	07	No lithium-polymer or non-rechargeable lithium batteries.	EuRoC-LV-RQT-0300
Parachute	CRAN-R2S-25-PAR	08	Accessible onboard batteries from ground.	EuRoC-LV-RQT-0310
Parachute	CRAN-R2S-25-PAR	09	Six hours battery lifetime on launch rail.	EuRoC-LV-RQT-0320
Parachute	CRAN-R2S-25-PAR	10	Mention non-parachute recovery methods in questionnaire.	EuRoC-LV-RQT-0330

Parachute	CRAN-R2S-25-PAR	11	Dual deployment recovery for apogee above 450 m.	EuRoC-LV-RQT-0340
Parachute	CRAN-R2S-25-PAR	12	Initial deployment event at/near apogee.	EuRoC-LV-RQT-0350
Parachute	CRAN-R2S-25-PAR	13	Initial deployment descent velocity 23-46 m/s.	EuRoC-LV-RQT-0360
Parachute	CRAN-R2S-25-PAR	14	Main deployment event below 450 m AGL.	EuRoC-LV-RQT-0370
Parachute	CRAN-R2S-25-PAR	15	Main deployment descent velocity <9 m/s.	EuRoC-LV-RQT-0380
Parachute	CRAN-R2S-25-PAR	16	Protect recovery system from hot ejection gases.	EuRoC-LV-RQT-0390
Parachute	CRAN-R2S-25-PAR	17	Use swivel links in recovery system rigging.	EuRoC-LV-RQT-0400
Parachute	CRAN-R2S-25-PAR	18	Dual deployment parachutes visually dissimilar.	EuRoC-LV-RQT-0410
Parachute	CRAN-R2S-25-PAR	19	Parachutes with clear contrast colours.	EuRoC-LV-RQT-0420
Parachute	CRAN-R2S-25-PAR	20	Use CATS Vega Flight Computer for altitude logging/tracking.	EuRoC-LV-RQT-0430
Parachute	CRAN-R2S-25-PAR	21	Assign call-sign to each CATS transmitter.	EuRoC-LV-RQT-0440
Parachute	CRAN-R2S-25-PAR	22	Fly specific firmware version in CATS flight computer.	EuRoC-LV-RQT-0450
Parachute	CRAN-R2S-25-PAR	23	Use CATS Ground Station for telemetry/tracking.	EuRoC-LV-RQT-0460
Parachute	CRAN-R2S-25-PAR	24	CATS devices comply with electronics requirements.	EuRoC-LV-RQT-0470
Parachute	CRAN-R2S-25-PAR	25	Implement cable management for safety critical wiring.	EuRoC-LV-RQT-0480
Parachute	CRAN-R2S-25-PAR	26	Secure safety critical wiring connections.	EuRoC-LV-RQT-0490

Parachute	CRAN-R2S-25-PAR	27	Use cryo-compatible insulation for wiring near cryo. devices.	EuRoC-LV-RQT-0500
Parachute	CRAN-R2S-25-PAR	28	Thermally test electronics and monitor temperature.	EuRoC-LV-RQT-0510
Parachute	CRAN-R2S-25-PAR	29	Ground test recovery system mechanisms.	EuRoC-LV-RQT-0520
Av-bay	CRAN-R2S-25-AVB	01	Implement redundant recovery system electronics.	EuRoC-LV-RQT-0240
Av-bay	CRAN-R2S-25-AVB	02	Use COTS flight computer for redundant recovery.	EuRoC-LV-RQT-0250
Av-bay	CRAN-R2S-25-AVB	03	Accessible electronics switches/connectors.	EuRoC-LV-RQT-0260
Av-bay	CRAN-R2S-25-AVB	04	Electronics accessible from ground.	EuRoC-LV-RQT-0270
Av-bay	CRAN-R2S-25-AVB	05	Mount electronics on side opposite to launch rail.	EuRoC-LV-RQT-0280
Av-bay	CRAN-R2S-25-AVB	06	No lithium-polymer or non-rechargeable lithium batteries.	EuRoC-LV-RQT-0300
Av-bay	CRAN-R2S-25-AVB	07	Accessible onboard batteries from ground.	EuRoC-LV-RQT-0310
Av-bay	CRAN-R2S-25-AVB	08	Six hours battery lifetime on launch rail.	EuRoC-LV-RQT-0320
Av-bay	CRAN-R2S-25-AVB	09	Use CATS Vega Flight Computer for altitude logging/tracking.	EuRoC-LV-RQT-0430
Av-bay	CRAN-R2S-25-AVB	10	Assign call-sign to each CATS transmitter.	EuRoC-LV-RQT-0440
Av-bay	CRAN-R2S-25-AVB	11	Fly specific firmware version in CATS flight computer.	EuRoC-LV-RQT-0450
Av-bay	CRAN-R2S-25-AVB	12	Use CATS Ground Station for telemetry/tracking.	EuRoC-LV-RQT-0460

Av-bay	CRAN-R2S-25-AVB	13	CATS devices comply with electronics requirements.	EuRoC-LV-RQT-0470
Av-bay	CRAN-R2S-25-AVB	14	Implement cable management for safety critical wiring.	EuRoC-LV-RQT-0480
Av-bay	CRAN-R2S-25-AVB	15	Secure safety critical wiring connections.	EuRoC-LV-RQT-0490
Av-bay	CRAN-R2S-25-AVB	16	Use cryo-compatible insulation for wiring near cryo. devices.	EuRoC-LV-RQT-0500
Av-bay	CRAN-R2S-25-AVB	17	Thermally test electronics and monitor temperature.	EuRoC-LV-RQT-0510
Av-bay	CRAN-R2S-25-AVB	18	Ground test recovery system mechanisms.	EuRoC-LV-RQT-0520
Nosecone	CRAN-R2S-25-NOC	01	Implement redundant recovery system electronics.	EuRoC-LV-RQT-0240
Nosecone	CRAN-R2S-25-NOC	02	Use COTS flight computer for redundant recovery.	EuRoC-LV-RQT-0250
Nosecone	CRAN-R2S-25-NOC	03	Accessible electronics switches/connectors.	EuRoC-LV-RQT-0260
Nosecone	CRAN-R2S-25-NOC	04	Electronics accessible from ground.	EuRoC-LV-RQT-0270
Nosecone	CRAN-R2S-25-NOC	05	Mount electronics on side opposite to launch rail.	EuRoC-LV-RQT-0280
Nosecone	CRAN-R2S-25-NOC	06	No lithium-polymer or non-rechargeable lithium batteries.	EuRoC-LV-RQT-0300
Nosecone	CRAN-R2S-25-NOC	07	Accessible onboard batteries from ground.	EuRoC-LV-RQT-0310
Nosecone	CRAN-R2S-25-NOC	08	Six hours battery lifetime on launch rail.	EuRoC-LV-RQT-0320
Nosecone	CRAN-R2S-25-NOC	09	Use CATS Vega Flight Computer for altitude logging/tracking.	EuRoC-LV-RQT-0430
Nosecone	CRAN-R2S-25-NOC	10	Assign call-sign to each CATS transmitter.	EuRoC-LV-RQT-0440

Nosecone	CRAN-R2S-25-NOC	11	Fly specific firmware version in CATS flight computer.	EuRoC-LV-RQT-0450
Nosecone	CRAN-R2S-25-NOC	12	Use CATS Ground Station for telemetry/tracking.	EuRoC-LV-RQT-0460
Nosecone	CRAN-R2S-25-NOC	13	CATS devices comply with electronics requirements.	EuRoC-LV-RQT-0470
Nosecone	CRAN-R2S-25-NOC	14	Implement cable management for safety critical wiring.	EuRoC-LV-RQT-0480
Nosecone	CRAN-R2S-25-NOC	15	Secure safety critical wiring connections.	EuRoC-LV-RQT-0490
Nosecone	CRAN-R2S-25-NOC	16	Use cryo-compatible insulation for wiring near cryo. devices.	EuRoC-LV-RQT-0500
Nosecone	CRAN-R2S-25-NOC	17	Thermally test electronics and monitor temperature.	EuRoC-LV-RQT-0510
Nosecone	CRAN-R2S-25-NOC	18	Ground test recovery system mechanisms.	EuRoC-LV-RQT-0520
Nosecone	CRAN-R2S-25-NOC	19	Use RF windows for internally mounted RF devices.	EuRoC-LV-RQT-0780
Nosecone	CRAN-R2S-25-NOC	20	RF windows must be 360° circ., two body calibers in length min.	EuRoC-LV-RQT-0790
Nosecone	CRAN-R2S-25-NOC	21	RF windows must not be made of carbon fibre.	EuRoC-LV-RQT-0800
Nosecone	CRAN-R2S-25-NOC	22	Keep RF antennas away from wiring and metallic elements.	EuRoC-LV-RQT-0810
Nosecone	CRAN-R2S-25-NOC	23	Place int. RF antennas at midpoint of RF window section.	EuRoC-LV-RQT-0820
Design & Verification	CRAN-R2S-25-DAV	01	Design and verification requirements for system components.	
Design & Verification	CRAN-R2S-25-DAV	02	Structural Design	
Design & Verification	CRAN-R2S-25-DAV	03	Verification standards	

Environmental	CRAN-R2S-25-ENV	01	General environmental standards
Function & Mission	CRAN-R2S-25-FAM	01	General environmental standards
Function & Mission	CRAN-R2S-25-FAM	02	Mission assurance requirements for payloads
Human Factor	CRAN-R2S-25-HUM	01	Human factor engineering and ergonomic requirements
Interface	CRAN-R2S-25-INT	01	System component interface standards
Operation	CRAN-R2S-25-OPR	01	General operation standards
Operation	CRAN-R2S-25-OPR	02	Tested launch abort/reset safety

A9 Heat Transfer Analysis

11.1.1 Introduction

The design of a rocket combustion chamber presents a significant thermal challenge due to the extremely high temperatures generated during fuel combustion. These thermal loads critically influence material selection, wall thickness, and the configuration of thermal protection systems. Effective thermal management is essential not only for maintaining structural integrity during operation but also for ensuring the overall reliability and efficiency of the propulsion system.

To address these challenges, this study adopts a two-tiered computational strategy to evaluate the thermal performance of composite chamber walls over the expected rocket burn duration. The chamber wall is modelled as a two-layer cylindrical structure comprising a low-conductivity insulation layer and a high-conductivity metallic casing.

The first tier involves a 1D transient radial heat transfer model developed in MATLAB using the explicit Finite Difference Method (FTCS scheme). This simplified model enables rapid prediction of temperature evolution through the wall layers, offering insight into thermal gradients and insulation effectiveness over a 7-second burn time. It assumes constant thermal properties—a valid simplification at short durations—and predicts a peak temperature of approximately **500 °C** at the metal casing's inner surface. Although this approaches the degradation

temperature of aluminium, the model excludes the additional insulating effect of the surrounding unburned fuel grain, which provides significant thermal resistance during combustion.

The second tier consists of a **transient CFD simulation** performed in STAR-CCM+. While it does not include fluid-structure interaction or thermal stress modelling, this simulation provides a detailed temperature distribution within the combustion chamber resulting from the high-temperature gas flow, modelled as hot air. Conducted over a **15-second physical time**, this analysis identifies critical thermal hotspots—particularly near the nozzle walls—where boundary layer effects locally increase static temperatures. These results guide insulation placement and support validation of the wall design under operating conditions.

Although thermal stresses have not been evaluated in this simulation, they are expected to be secondary to mechanical hoop stresses, for which the chamber is structurally designed. Thus, this study focuses on thermal behaviour as the primary design criterion. However, the **transient CFD simulation lays the foundation for future high-fidelity CHT-FEA analysis**, where both thermal and mechanical effects will be coupled to assess the full thermo-mechanical performance during the burn.

This integrated approach enables early validation of the design for a given burn time and serves as an **optimum method for critical design review**, identifying areas of potential overheating and enabling design refinement before complex structural simulations.

11.1.2 Transient Heat Transfer Analysis Using MATLAB

1.Mathematical Model and Numerical Methodology

The transient heat conduction in the radial direction of a composite hollow cylinder is governed by the 1D heat equation in cylindrical coordinates:

$\partial T/\partial t = \alpha(\partial^2/\partial r^2(T) + 1/r\ \partial/\partial r(T))$ ----------Governing Differential Equation

Where α = thermal diffusivity of the material

To solve this PDE numerically, the explicit Forward Time Central Space (FTCS) finite difference method is implemented. The domain is discretized into uniform radial segments, and time is advanced explicitly using:

$T,_i^{(n+1)} = T,_i^n + d\ (T,^n_{i+1} - 2T,^n_i + T,^n_{i-1}) + d1/ri\ (T,^n_{i+1} - T,^n_{i-1})$----------Discretised Differential Equation

Where,

$d = \alpha\ \Delta T / \Delta r^2$ = diffusion number

$d1 = \alpha\ \Delta T / 2\Delta r$

2. Numerical Stability Consideration

Given the use of an explicit FTCS scheme, numerical stability is a critical concern. The scheme is conditionally stable, requiring:

$\Delta T <= ½ \Delta r^2 / \alpha,max$

where αmaxis the highest thermal diffusivity among the layers. The code dynamically computes this maximum allowable time step and adjusts the user-defined burn time to avoid instability. This ensures accurate and stable simulations throughout the prescribed burn duration.

3. Boundary and Initial Conditions

The boundary conditions are applied as follows:

- **Inner Surface (r = r_in):** Convective boundary condition to simulate hot gas exposure from the rocket core. The inner surface exchanges heat with a fluid at 3000 K through a convection coefficient of 430 W/m²K, computed from the equation:

  ((C*Cp_mix*G^0.8)/(D_internal)^0.2)*(1+(D_internal/length_chamber)^0.7) (1)

  Where,

  C = 3.075 is constant

  Cp_mix = 2.54 is the specific heat of mixture at O/F 2.1 (kJ/kgK)

  w = w_propellant/t_burn, mass flow rate of propellent (kg/s)

  w_propellant = weight of the propellant in kg = 7.5 kg

  t_ burn = 5s = burn time in seconds

  S =pi/4*(D_internal)^2 is the cross-sectional area (m^2)

  G = w/S; is the mass velocity (kg/sec-m^2)

  Using the above formula we get rate of convection heat transfer = 1570 W/m^2K

- **Outer Surface (r = r_out):** Dirichlet boundary condition, fixed at 300 K to model ideal heat sinking or thermal clamping.

Initial temperature across the entire wall is set to 300 K, representing pre-burn ambient conditions.

4. Material Properties and Assumptions

The model treats each layer with constant thermal conductivity and specific heat, despite their known temperature dependency. This assumption is valid for the short burn duration of interest (e.g., 5 seconds), during which the properties do not vary drastically. The inner insulation layer has significantly lower thermal conductivity and heat capacity than the metallic casing, leading to delayed heat penetration.

5. Output Interpretation and Applications

The code provides two essential outputs:

- **Radial Temperature Profile:** Temperature across the cylinder wall at the final time step, indicating how effectively the insulation layer resists thermal penetration.

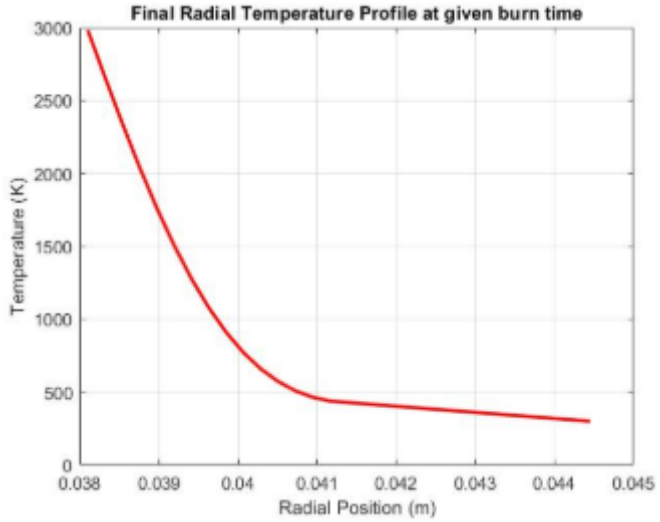


Figure 21:Temperature variation across chamber at burn time.

In burn time of 7 seconds, the outer casing of metal is at nearly 300K that is ambient temperature, which suggest the temperature has not penetrated from the casing.

- **Temporal Variation at Target Radius:** Temperature evolution at a user-defined radius, which is the inner radius of metal casing which is subjected to much higher temperature.

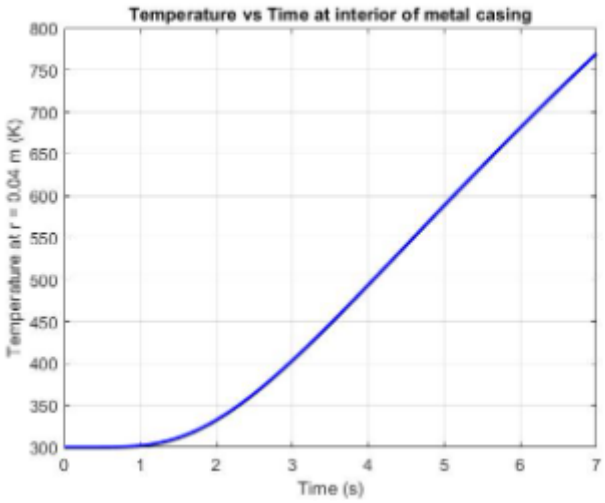


Figure 22:Temperature variation with respect to time at inner radius of metal casing

As we can see from the plot in 8 seconds the temperature at the inner section of metal casing gets close to 770 K which is close to 500 deg Celsius, however this model does not take into account the added protection from the fuel grain so even for a conservative model the design is safe.

6. Justification for Explicit Scheme

Although implicit schemes are unconditionally stable and allow larger time steps, the explicit FTCS method was preferred for this problem due to:

- **Simplicity and Transparency:** Easy to implement and modify for multilayer materials.
- **Short Simulation Time:** Given the relatively small burn time and fine discretization, the explicit method runs efficiently without excessive computational cost.
- **Fine Temporal Resolution:** Provides high-resolution temporal data, important for capturing rapid thermal transients.

11.1.3 3D CFD Analysis

To assess the thermal integrity of the combustion chamber, a full 3D CFD analysis was carried out using STAR-CCM+. The aim was to identify the regions where maximum temperature is expected.

Simulation Setup:

- **Geometry:** 3D combustion chamber and nozzle.

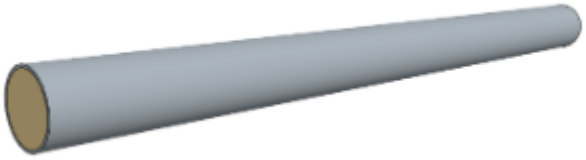


Figure 23: 3D-CAD of Combustion chamber and nozzle

- **Mesh:** Polyhedral mesh for the fluid domain

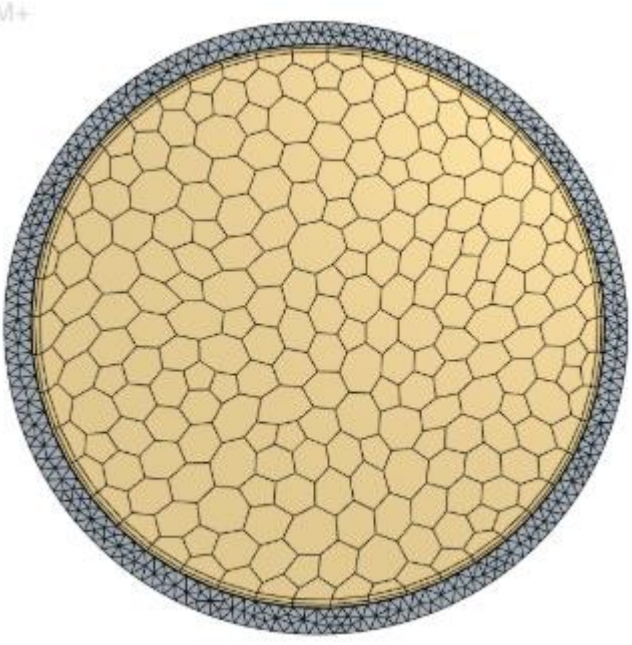


Figure 24:Polyhedral Mesh

- **Solver Settings:**

Turbulence Model: K-epsilon 2 equation

Implicit unsteady solver with the following specification:

Time step = 10^{-6} sec

Maximum inner iteration = 25

Maximum physical time = 15 sec

Maximum steps = 5000

These settings were optimized through iterative testing to ensure numerical stability and solution accuracy. Initial trials using the default time step of 1×10^{-3} s led to divergence, with residuals escalating uncontrollably due to the inability of the solver to capture rapid thermal transients. Reducing the time step to 1×10^{-6} s, combined with an increased number of inner iterations per time step, allowed the energy equation to converge consistently within each time increment. Although this configuration significantly increased the computational load—requiring nearly 50,000 iterations to resolve the full thermal boundary layer—it ensured a stable and physically accurate solution critical for assessing transient thermal and structural responses of the chamber wall.

Assumptions:

- Reacting flow replaced by hot air of equivalent temperature and velocity.
- Simplifies modelling of complex multi-phase combustion without compromising thermal evaluation.
- Physical time of simulation is taken to 15 seconds instead of 7 seconds of targeted burn time to make the simulation more conservative.
- Due to the non-linear stress-strain behaviour of phenolic insulation, the simulation does not incorporate the insulation. This also makes the simulation further more conservative

Key Outputs:

- Temperature contours inside of chamber



Figure 25: Temperature within the combustion chamber fluid domain

Due to the development of the boundary layers downstream of the flow, flow locally decelerates which increases the static temperature. This region of overheating is identified through this CFD simulation.

11.1.4 Conclusion

This study presents an integrated thermal evaluation strategy for rocket combustion chamber walls, combining a simplified 1D transient heat conduction model with a detailed **transient CFD simulation**. The MATLAB model quickly assessed temperature rise through the wall layers, validating the insulation's performance under the expected 7-second burn duration. It highlighted a critical temperature approaching material limits but remained within acceptable bounds, assuming the presence of additional insulating effects from the unburned fuel grain.